

A
INDUSTRIAL ORIENTED MINI PROJECT
REPORT
ON
SMART IRRIGATION SYSTEM

ABSTRACT

SMART IRRIGATION SYSTEM

The ever-growing demand for sustainable agriculture necessitates the integration of modern technology into traditional farming practices. This project presents a wireless sensor Network (WSN)-based Smart Agriculture Monitoring and Controlling System, aimed at improving crop productivity, optimizing resource utilization, and minimizing human intervention.

At the core of the system is the Arduino Uno microcontroller, which acts as the central processing unit. It collects real-time data from multiple sensors deployed in the agricultural field. The DHT11 sensor is used to measure environmental parameters such as temperature and humidity, while a soil moisture sensor monitors the water content present in the soil. These sensors continuously transmit data to the microcontroller for analysis.

The collected data is displayed on an 12C LCD display, enabling farmers to easily monitor field conditions in real-time. This visual feedback helps in making informed decisions regarding irrigation and crop management without the need for manual field inspection.

A key feature of this system is its automated irrigation control mechanism. The system incorporates relay modules connected to water pump motors. When the soil moisture level falls below a predefined threshold, the Arduino automatically activates the relay, turning on the water pump to irrigate the field. Once the required moisture level is restored, the pump is switched off, ensuring efficient water usage. Additionally, the system can be programmed to respond to temperature variations—activating cooling mechanisms or irrigation when the temperature exceeds a critical limit, thereby protecting crops from heat stress.

The integration of Wireless Sensor Network (WSN) technology allows data transmission over a distance, making it suitable for remote monitoring. This can be further enhanced by integrating IoT platforms, enabling farmers to access real-time data through smartphones or web applications from anywhere. This smart agriculture solution significantly reduces water and energy wastage by applying resources only when necessary. It promotes precision agriculture, where inputs like water and environmental control are optimized based on real-time data. The system is designed to be cost-effective, scalable, and user-friendly, making it ideal for small-scale farmers as well as large agricultural fields.

INDEX

S.NO	TITLE	PAGE NO.
1.	Introduction	1-2
	Traditional Cost Comparison Methods	1
	Motivation	1-2
	Challenges in Previous Cost Comparison Approaches	2
	Objectives of Cost Compass	2
2.	Literature Survey	3-4
	2.1 Wireless Sensor Networks (WSN) in Agriculture	3
	2.2 Automation and Control in Agriculture	3-4
	2.3 Sensor Technologies in Agriculture	4
	2.4 Challenges and Future Directions	4
3.	Feasibility Study	5-8
	3.1 Technological Feasibility	5
	3.1.1 Technologies Involved	5
	3.1.2 Scalability and Performance	5-6
	3.1.3 Risk Management	6
	3.2 Operational Feasibility	6
	3.2.1 System Operations	7
	3.2.2 Staffing and Resources	7
	3.2.3 User Training	7-8
	3.2.4 Maintenance and Support	8
	3.2.5 Cost and Resource Allocation	8
4.	System Requirements	9-19
	4.1 Existing System	9
	4.2 Proposed System	9-11
	4.3 Software Requirements	11
	4.4 Hardware Requirements	11-19

5.	System Design	20-22
5.1	System Architecture	20
5.2	Source code	21-22
6.	Software Implementation	23-34
6.1	Software development	23
7.	System Testing	35-37
7.1	Unit Testing	35
7.2	Integration Testing	35
7.3	System Testing	36
7.4	Acceptance Testing	36
7.5	Test Cases for Online System	36-37
8.	Results and Output Screens	38-41
8.1	Initial User Interface	38
8.2	System Setup	39
8.3	Sensor Data Display	40
8.4	Final output Display	41
9.	Conclusion	42
10.	Future Enhancement	43
11.	References	44-45

LIST OF FIGURES

S. NO	TITLE	PAGE NO.
4.1	Block Diagram of Smart Irrigation System	11
4.2	Arduino UNO of Smart Irrigation	12
4.3	Pinout Diagram for Arduino MKR1000 WIFI	13
4.4	Power Supply Circuit Diagram	14
4.5	Filter Capacitor Circuit	16
4.6	Voltage Regulator Circuit	17
4.7	LCD Display Module	18
4.8	LCD Interfacing Diagram	18
4.9	Soil Moisture Sensor	18
4.10	DHT11 Temperature and Humidity Sensor	19
4.11	Relay Module Circuit	19
5.1	Smart Irrigation System Architecture	20
6.1	Arduino IDE Program Output	23
6.2	LED Blink Program Output	26
6.3	Switch Circuit Using Arduino	31
8.1	System Initialization Display	38
8.2	Complete Hardware Setup	39
8.3	Real-Time Sensor Data Display	40
8.4	Final System Output Display	41

CHAPTER 1

INTRODUCTION

1. Introduction

Agriculture is one of the most important sectors, and efficient use of water is essential for better crop production. In traditional irrigation methods, water is supplied manually without considering the actual needs of the soil. This often leads to overwatering or underwatering, resulting in water wastage and reduced crop quality.

To overcome these problems, the Smart Irrigation System is introduced. This system uses sensors to monitor soil moisture, temperature, and humidity, and automatically controls the irrigation process. When the soil becomes dry, the system turns on the water pump, and when sufficient moisture is reached, it turns it off.

The main aim of this system is to save water, reduce human effort, and improve agricultural productivity. It is a cost-effective and efficient solution that can be easily used by farmers. By using modern technology like microcontrollers and sensors, the system ensures smart and precise irrigation.

1.1 Traditional Irrigation Methods

Traditional irrigation methods such as manual watering, flood irrigation, and sprinkler systems have been widely used in agriculture for many years, but they come with several limitations that affect efficiency and resource management. Manual irrigation requires continuous human effort and regular monitoring of the fields, making it time-consuming and less efficient, especially for large-scale farming. Flood irrigation often leads to excessive water usage and uneven distribution, where some areas receive more water than needed while others remain under-watered. Sprinkler systems, although an improvement over manual methods, still lack the ability to make real-time decisions based on actual soil and environmental conditions. Similarly, time-based irrigation systems supply water at fixed intervals without considering the current moisture level of the soil, which can result in over-irrigation or under-irrigation. Overall, these traditional approaches do not take into account real-time environmental changes, leading to inefficient water usage and highlighting the need for smarter, automated irrigation solutions.

1.2 Motivation

The main motivation behind developing the Smart Irrigation System is to address the problems associated with traditional irrigation methods.

- Increasing water scarcity demands efficient usage

- Need to reduce manual labour in agriculture
- Requirement for real-time monitoring and automation
- Desire to improve crop productivity and quality
- Adoption of modern technologies in farming practices

The system aims to provide a smart and efficient irrigation solution that benefits farmers and conserves natural resources.

1.3 Challenges in Previous Irrigation Approaches

Existing irrigation methods and basic automated systems face several challenges:

Lack of Real-Time Monitoring:

Most systems do not adjust irrigation based on current soil conditions.

Water Wastage:

Over-irrigation leads to unnecessary water consumption.

High Labour Dependency:

Farmers need to manually monitor and control irrigation.

Limited Automation:

Traditional systems lack intelligent decision-making capabilities.

Cost Issues:

Advanced irrigation systems can be expensive and not affordable for small farmers.

Maintenance Difficulties:

Some systems require complex maintenance and technical knowledge.

These challenges highlight the need for a simple, cost-effective, and automated irrigation solution.

1.4 Objectives of Smart Irrigation System

The primary objectives of the Smart Irrigation System are:

To automate irrigation based on soil moisture levels

To reduce water wastage and promote efficient usage

To minimize human intervention in irrigation processes

To provide real-time monitoring of environmental conditions

To develop a cost-effective and easy-to-use system

To improve crop yield and agricultural productivity

To enable future enhancements like IOT- based remote monitoring

CHAPTER 2

LITERATURE SURVEY

The integration of Wireless Sensor Networks (WSN) agriculture has gained significant attention due to the advantages they offer in terms of monitoring, controlling, and automating various agricultural processes. This section presents an overview of the existing research and development in the area of smart agriculture and related technologies, highlighting the use of ARDUINO UNO-based automation, and sensors for real-time monitoring and control of environmental parameters such as soil moisture, temperature, humidity, and irrigation systems.

2.1 Wireless Sensor Networks (WSN) in Agriculture

Wireless Sensor Networks (WSNs) are fundamental to modern precision agriculture systems. These networks involve multiple sensors placed in the field, which wirelessly transmit data to a central system or cloud platform for analysis and decision-making. WSNs are particularly useful in precision farming which aims to optimize agricultural production through detailed, real-time monitoring of environmental conditions.

Sundararajan et al. (2018) explored the use of WSNs for agricultural monitoring systems. They designed a system that incorporated soil moisture sensors, temperature sensors, and humidity sensors to monitor the conditions of agricultural fields. Their system used an ARDUINO UNO microcontroller for data collection and control, highlighting the role of WSNs in ensuring optimal resource management in farming. The authors concluded that WSNs, when integrated with automation systems, provide significant benefits in terms of resource efficiency and yield optimization.

2.2 Automation and Control in Agriculture

The automation of agricultural processes, such as irrigation and cooling, plays a critical role in improving the efficiency and sustainability of farming practices. Automated systems based on sensor inputs help farmers reduce manual labour and ensure that the agricultural processes are executed with high precision.

Alcocer et al. (2015) presented an automated irrigation system based on soil moisture sensors and ARDUINO UNO microcontrollers. The system was designed to irrigate crops automatically when the soil moisture level dropped below a predefined threshold. They also integrated relay-controlled water pumps for irrigation and reported significant water savings by reducing over-irrigation. This study demonstrated the feasibility of automated systems for irrigation in precision agriculture, emphasizing the role of sensors and automation in improving resource use efficiency.

In another study, Agra et al. (2019) explored the use of ARDUINO UNO-controlled motors to activate irrigation systems and cooling systems based on temperature and soil conditions. Their system incorporated temperature sensors to regulate the cooling process in greenhouses, preventing crops from overheating. The authors concluded that automation systems based on ARDUINO UNO offered an effective solution for climate management and energy efficiency in agriculture.

2.3 Sensor Technologies in Agriculture

The use of sensors for environmental monitoring is fundamental to modern agriculture. Sensors allow for precise measurement of environmental conditions, and the data they provide can be used to inform decision-making processes and trigger automated systems.

DHT-11, a commonly used temperature and humidity sensor, has been widely employed in various agricultural monitoring systems. According to Rafiq et al. (2017), the DHT-11 sensor was used to monitor temperature and humidity levels in greenhouse environments. The sensor was integrated with ARDUINO UNO microcontrollers to activate cooling systems when temperatures exceeded the desired thresholds, reducing the risk of crop damage due to heat stress.

Additionally, soil moisture sensors are integral to precision farming. González et al. (2016) explored the use of soil moisture sensors for monitoring irrigation systems in crops. By accurately measuring soil moisture, farmers can optimize irrigation, thus conserving water and reducing costs. The integration of soil moisture sensors with automated irrigation systems ensures that crops receive the right amount of water, leading to improved crop health and higher yields.

2.4 Challenges and Future Directions

While significant advancements have been made in the development of smart agriculture technologies, there are still challenges that need to be addressed. Connectivity issues, particularly in rural areas, can limit the effectiveness of agricultural systems. In addition, the cost of sensors and equipment may be a barrier to adoption for small-scale farmers. The long-term sustainability of these systems depends on cost-effective solutions and easy scalability.

Future research should focus on improving sensor accuracy and developing low-cost solutions for small-scale farmers. Additionally, integrating artificial intelligence and machine learning with agricultural monitoring systems can enable more sophisticated decision-making based on historical data and predictive models.

CHAPTER 3

FEASIBILITY STUDY

3.1 Technological Feasibility

Technological feasibility analyzes whether the required technology is available, affordable, and suitable for developing the Smart Irrigation System. The proposed system uses simple hardware components and embedded programming, making it highly feasible even for beginners and small-scale applications. The system is built using microcontroller-based automation, where sensors collect environmental data and the controller processes it to take necessary actions. These technologies are widely used in agriculture and IoT-based systems, ensuring reliability and effectiveness.

3.1.1 Technologies Involved

The Smart Irrigation System integrates both hardware and software technologies to achieve automation, efficiency, and ease of operation in agricultural practices. At the core of the system is the Arduino Uno, which functions as the main controller, responsible for collecting data from sensors and making decisions based on predefined conditions. The soil moisture sensor plays a crucial role by measuring the water content in the soil, helping the system determine whether irrigation is required. Alongside this, the DHT-11 sensor monitors environmental parameters such as temperature and humidity, which can influence irrigation needs and overall crop health. The relay module acts as a switching device, enabling the Arduino to control the water pump by turning it ON or OFF automatically based on sensor data. The water pump is used to supply water to the field whenever the soil moisture level drops below the set threshold, ensuring timely irrigation. An I2C LCD display is incorporated to provide real-time information, showing sensor readings like temperature, humidity, and soil condition, making it easy for users to monitor system performance. The entire system is powered by a suitable power supply that ensures stable operation of all components. The behavior of the system is programmed using Embedded C, which defines how the Arduino processes inputs and controls outputs. Overall, these technologies are widely available, cost-effective, and simple to implement, making the system practical and accessible for farmers with minimal technical knowledge.

3.1.2 Scalability and Performance

The Smart Irrigation System is designed not only for efficient operation but also with future expansion and adaptability in mind. In terms of scalability, the system can be easily extended by adding more sensors to monitor larger agricultural areas, making it suitable for both small

farms and large fields. It can also be integrated with IoT platforms to enable remote monitoring and control through mobile devices or web applications. The system demonstrates strong performance by providing real-time monitoring of environmental conditions and responding quickly to changes in soil moisture levels, ensuring timely irrigation and preventing over-watering. Its flexibility allows for further enhancements, such as incorporating weather prediction systems, mobile app connectivity, and cloud-based data storage for advanced analysis and decision-making. Additionally, the system is energy efficient, consuming minimal power, which makes it ideal for rural or remote areas with limited electricity availability. Overall, the system delivers reliable and consistent performance while offering the ability to scale and adapt according to the evolving needs of modern agriculture.

3.1.3 Risk Management

Although the Smart Irrigation System is practical and efficient, certain risks must be carefully considered and managed to ensure reliable operation. One major concern is hardware failure, as sensors and electronic components may degrade over time due to continuous usage or environmental exposure; this can be minimized through regular maintenance and timely replacement of faulty parts. Power supply issues are another potential risk, since voltage fluctuations or power outages can disrupt system performance; using voltage regulators, stabilizers, or backup power sources like batteries or solar systems can help address this problem. Environmental conditions such as extreme heat, humidity, or rainfall may also affect the accuracy and durability of sensors, making proper casing and protective measures essential. Additionally, data accuracy is critical, as incorrect sensor readings due to poor calibration can lead to improper irrigation decisions; therefore, periodic calibration and testing of sensors are necessary. There is also the risk of water pump malfunction, which could interrupt irrigation at crucial times; implementing backup systems or alert mechanisms can help mitigate this issue. By identifying and addressing these risks in advance, the system can be made more robust, reliable, and suitable for real-world agricultural applications.

3.2 Operational Feasibility

Operational feasibility determines whether the system can be effectively used in real-world conditions. The Smart Irrigation System is designed to be user-friendly and suitable for farmers with minimal technical knowledge. It reduces manual effort and ensures efficient irrigation, making it highly practical for agricultural use.

3.2.1 System Operations

The Smart Irrigation System operates automatically with minimal human intervention, making it highly efficient and user-friendly. In this system, sensors continuously monitor important environmental parameters such as soil moisture, temperature, and humidity in real time. The collected data is sent to the Arduino controller, which acts as the central processing unit. The controller analyzes the incoming data and compares it with predefined threshold values programmed into the system. Based on this analysis, appropriate actions are taken automatically; for instance, if the soil moisture level falls below the required limit, the system activates the water pump to irrigate the field. Once the soil reaches sufficient moisture, the pump is turned OFF to prevent over-irrigation. Throughout this process, real-time sensor data and system status are displayed on the LCD screen, allowing users to easily monitor operations. This automated workflow not only reduces the need for manual effort but also ensures efficient water usage, ultimately helping to improve crop productivity and support sustainable farming practices.

3.2.2 Staffing and Resources

The Smart Irrigation System is designed to operate with minimal human resource requirements, making it highly practical and accessible for farmers. It does not require highly skilled professionals to manage or operate the system, as it is simple and user-friendly. With basic training, farmers can easily understand how to use and monitor the system effectively. Due to its automated nature, tasks such as irrigation and environmental monitoring are handled by the system itself, reducing the need for constant supervision. Additionally, the components used, such as sensors and microcontrollers, are widely available and easy to install and maintain. This low dependency on human effort and technical expertise makes the system suitable for both small-scale farmers with limited resources and large-scale agricultural operations aiming for efficiency and productivity.

3.2.3 User Training

The Smart Irrigation System is designed to be simple, user-friendly, and easy to operate, ensuring that it can be used by individuals with minimal technical knowledge. Users only need basic understanding, such as how to switch the system ON and OFF and how to read the information displayed on the LCD screen, including temperature, humidity, and soil moisture levels. The system interface is straightforward, making it easy for farmers to interpret the data and understand the system's functioning. Additionally, users can learn basic troubleshooting steps, such as checking connections or ensuring proper power supply, to handle minor issues.

The training required to operate the system is minimal and can be provided within a short period, making it highly accessible even to those without a technical background. This ease of use ensures that the system can be widely adopted by farmers, improving efficiency and productivity without adding complexity.

3.2.4 Maintenance and Support

Regular maintenance is essential to ensure the long-term efficiency and reliability of the Smart Irrigation System. Periodic cleaning of the soil moisture sensor is required to remove dirt and deposits that may affect its accuracy and lead to incorrect readings. It is also important to check wiring connections regularly to ensure there are no loose or damaged connections that could interrupt system operation. Any faulty or worn-out components should be replaced promptly to maintain smooth functioning. Additionally, the performance of the water pump should be monitored to ensure it operates efficiently without any blockages or mechanical issues. Since the overall system design is simple and uses easily available components, maintenance tasks are straightforward, require minimal technical expertise, and are cost-effective, making the system practical for long-term use in agricultural environments.

3.2.5 Cost and Resource Allocation

The Smart Irrigation System is economically feasible and offers a cost-effective solution for modern agriculture. It is built using low-cost components such as the Arduino microcontroller and basic sensors, which are easily available in the market at affordable prices. The installation cost of the system is also minimal, as it does not require complex infrastructure or expensive equipment. By automating irrigation based on real-time soil moisture levels, the system significantly reduces water wastage and unnecessary electricity consumption, leading to lower operational costs over time. This efficient use of resources not only saves money but also increases overall productivity. Considering the low initial investment and the long-term benefits, the system proves to be highly affordable and practical, especially for farmers who seek to improve efficiency without incurring high expenses.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 EXISTING SYSTEM

The current agricultural practices primarily depend on traditional methods, where farmers manually monitor and manage various environmental parameters such as soil moisture, temperature, humidity, and irrigation schedules. These methods are time-consuming, labour-intensive, and often prone to human error.

Traditional farming methods face several challenges that limit efficiency and productivity. One major issue is manual monitoring, where farmers need to frequently visit fields to check soil moisture and environmental conditions, which is time-consuming and difficult, especially for large agricultural areas. Another significant problem is water wastage, as irrigation is often carried out based on fixed schedules rather than actual soil needs, leading to either over-watering or under-watering of crops. This inefficient use of water is particularly concerning in regions with limited water resources.

Additionally, there is energy inefficiency, since the absence of automated systems forces farmers to manually manage temperature conditions, often resulting in unnecessary energy consumption. Traditional systems also lack real-time data, making it hard for farmers to make quick and informed decisions regarding irrigation, crop health, or environmental changes.

Furthermore, limited automation increases dependency on manual labour for routine tasks like irrigation, which not only adds to the workload but also reduces overall productivity. These challenges highlight the need for smart and automated agricultural solutions.

4.2 PROPOSED SYSTEM

The WSN-Based Smart Agriculture Monitoring and Controlling System aims to overcome the limitations of traditional farming methods by integrating Wireless Sensor Networks (WSN), ARDUINO UNO-based automation, and sensor-based control systems.

The proposed system provides an intelligent solution for real-time monitoring and automated control of various environmental parameters, such as soil moisture, temperature, and humidity. Here are the key features and advantages of the proposed system:

Key Features:

The Smart Agriculture Monitoring and Controlling System is designed with key features that enhance efficiency and sustainability in farming practices. The system performs real-time monitoring using a DHT-11 sensor for measuring temperature and humidity, along with a soil moisture sensor to assess the water content in the soil. These sensor readings are continuously updated and displayed on an LCD screen, allowing users to easily observe current environmental conditions. Based on this data, the system enables automated irrigation, where the water pump is controlled through a relay module. When the soil moisture level drops below a predefined threshold, the pump is automatically activated to supply water, and it is turned off once the desired moisture level is achieved, thereby preventing water wastage. Additionally, the system supports wireless communication, allowing sensor data to be transmitted over a network without the need for complex wiring, making it flexible and scalable for different agricultural environments. Another important aspect is energy and resource efficiency, as the system minimizes unnecessary usage of water and power by operating only when required. This not only reduces operational costs but also promotes sustainable farming, especially in regions facing water scarcity. Overall, these features work together to provide an intelligent, reliable, and efficient solution for modern agriculture.

Advantages of the Proposed System:

The Smart Agriculture Monitoring and Controlling System offers several important advantages that make it highly beneficial for modern farming. One of the key benefits is increased productivity, as the automation of irrigation, cooling, and environmental monitoring ensures that crops receive the right amount of water and suitable conditions at the right time, leading to improved growth and higher yields. The system also significantly reduces labour requirements, since tasks like irrigation are performed automatically based on real-time sensor data, minimizing the need for constant human supervision. In addition, the system promotes sustainability by optimizing the use of water and energy, helping to conserve valuable resources and support environmentally friendly farming practices, which is especially important in areas facing water scarcity. Another major advantage is its cost-effectiveness, as it uses affordable components such as the DHT-11 sensor, soil moisture sensor, and Arduino Uno microcontroller, making it accessible even for small-scale farmers. Furthermore, the system is highly scalable, meaning it can be easily adapted for use in different agricultural settings, including small farms, large fields, and greenhouses. Overall, these advantages make the system a practical, efficient, and sustainable solution for improving agricultural productivity.

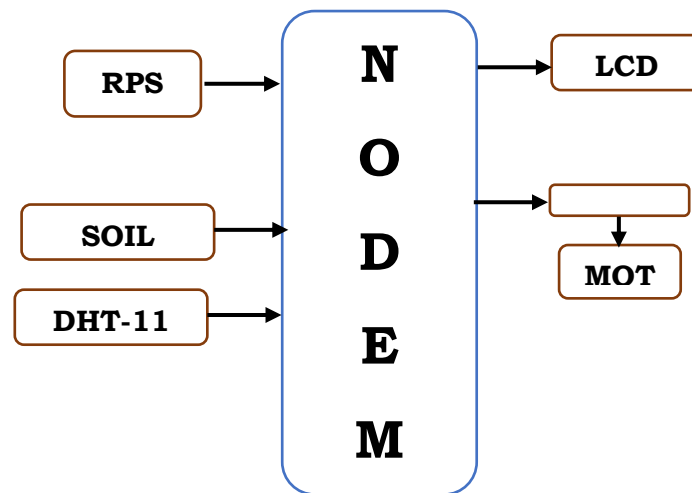


Figure 4.1: Block Design Proposal for the System

4.3 Software Requirements

The Smart Irrigation System mainly uses embedded programming and simple development tools rather than complex web-based software. The software is responsible for reading sensor data, processing it, and controlling the irrigation system. The Smart Irrigation System uses Embedded C programming as its backend logic to control and manage all operations efficiently. The code is written, compiled, and uploaded to the Arduino Uno using the Arduino IDE, which provides a simple and user-friendly platform for development. The Arduino functions as the central processing unit of the system, where all decision-making processes take place. It continuously receives data from sensors such as the soil moisture sensor and DHT-11 sensor, processes this information, and compares it with predefined threshold values set in the program. Based on these conditions, the Arduino automatically controls the output devices, such as turning the water pump ON or OFF through the relay module. This programming logic ensures that the system operates accurately and responds quickly to changing environmental conditions, enabling efficient automation and reliable performance in agricultural applications.

4.4 Hardware Requirements

General description of ARDUINO

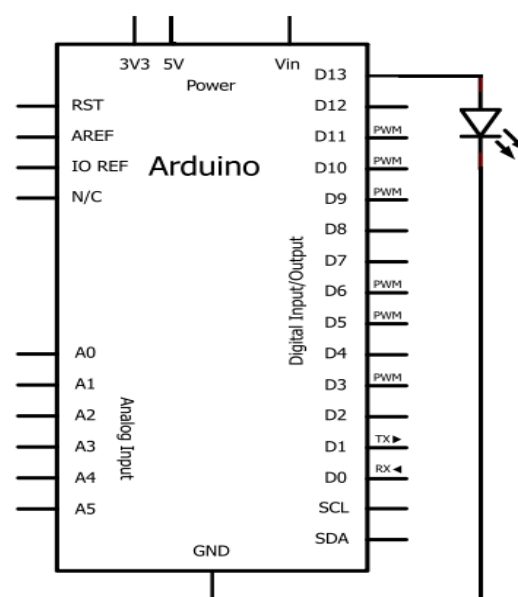
Arduino is an open source computer hardware and software company, project, and user community that manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical and digital world. The project's products are distributed as open-source hardware and software, which are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL),[1] permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or Breadboards (shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.



Figure 4.2: Arduino UNO of Smart Irrigation

Pin diagram:



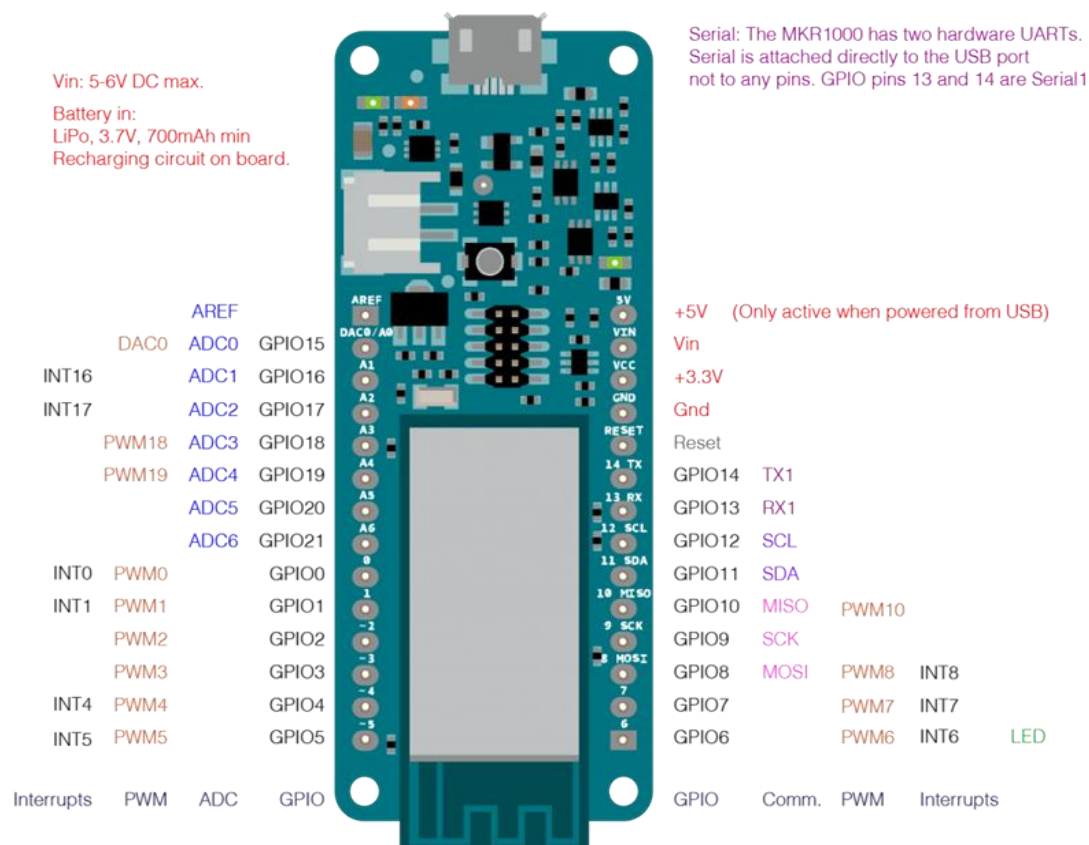


Figure 4.3: Pinout Diagram For The Arduino MKR1000 WIFI

Pin Description

Pin Category	Pin Name	Details
Power	Vin, 3.3V, 5 V, GND	Vin: Input Voltage to Arduino. Whe using an external Power source. 5V: Regulated power supply used to power microcontroller and other components on the board. 3.3V: 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA. GND: Ground Pins.

Reset	Reset	Resets the Microcontroller.
Analog Pins	A0-A5	Used to provide analog input in the range of 0-5 V
Input/Output Pins	Digital Pins 0-13	Can be used as input or output pins.
Serial	0(Rx), 1(Tx)	Used to receive and transmit TTL serial data.
External Interrupts	2.3	To trigger an interrupt.
PWM	3,5,6,9,11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) AND 13 (SCK)	Used for SPI communication
Inbuilt LED	Digital Pins 0 – 13	To turn on the in-built LED
TWI	A4 (SDA), A5 (SCA)	Used for TWI communication
AREF	AREF	To provide reference voltage for input voltage.

Power Supply

All digital circuits require regulated power supply. In this article we are going to learn how to get a regulated positive supply from the mains supply.

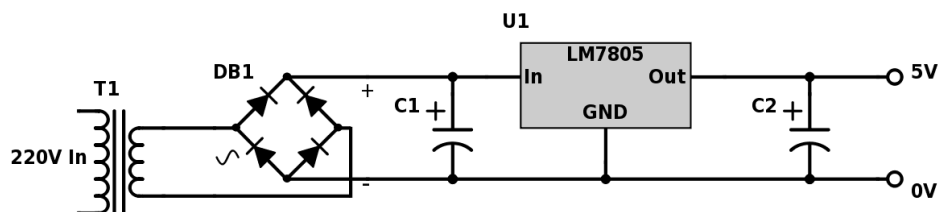


Figure 4.4: Power Supply Circuit Diagram

TRANSFORMER

A transformer consists of two coils also called as “WINDINGS” namely PRIMARY & SECONDARY.

They are linked together through inductively coupled electrical conductors also called as CORE. A changing current in the primary causes a change in the Magnetic Field in the core & this in turn induces an alternating voltage in the secondary coil. If load is applied to the secondary then an alternating current will flow through the load. If we consider an ideal condition then all the energy from the primary circuit will be transferred to the secondary circuit through the magnetic field.

$$P_{\text{primary}} = P_{\text{secondary}}$$

Rectifier

A rectifier is a device that converts an AC signal into DC signal. For rectification purpose we use a diode, a diode is a device that allows current to pass only in one direction i.e. when the anode of the diode is positive with respect to the cathode also called as forward biased condition & blocks current in the reversed biased condition.

Rectifier can be classified as follows:

1) Half Wave rectifier

This is the simplest type of rectifier as you can see in the diagram a half wave rectifier consists of only one diode. When an AC signal is applied to it during the positive half cycle the diode is forward biased & current flows through it. But during the negative half cycle diode is reverse biased & no current flows through it. Since only one half of the input reaches the output, it is very inefficient to be used in power supplies.

2) Full wave rectifier

Half wave rectifier is quite simple but it is very inefficient, for greater efficiency we would like to use both the half cycles of the AC signal. This can be achieved by using a center tapped transformer i.e. we would have to double the size of secondary winding & provide connection to the center. So during the positive half cycle diode D1 conducts & D2 is in reverse biased condition. During the negative half cycle diode D2 conducts & D1 is reverse biased. Thus we get both the half cycles across the load.

One of the disadvantages of Full Wave Rectifier design is the necessity of using a center tapped transformer, thus increasing the size & cost of the circuit. This can be avoided by using the Full Wave Bridge Rectifier.

3) Bridge Rectifier

As the name suggests it converts the full wave i.e. both the positive & the negative half cycle into DC thus it is much more efficient than Half Wave Rectifier & that too without using a center tapped transformer thus much more cost effective than Full Wave Rectifier.

Full Bridge Wave Rectifier consists of four diodes namely D1, D2, D3 and D4. During the positive half cycle diodes D1 & D4 conduct whereas in the negative half cycle diodes D2 & D3 conduct thus the diodes keep switching the transformer connections so we get positive half cycles in the output.

If we use a center tapped transformer for a bridge rectifier we can get both positive & negative half cycles which can thus be used for generating fixed positive & fixed negative voltages.

Filter Capacitor

Even though half wave & full wave rectifier give DC output, none of them provides a constant output voltage. For this we require to smoothen the waveform received from the rectifier. This can be done by using a capacitor at the output of the rectifier this capacitor is also called as “FILTER CAPACITOR” or “SMOOTHING CAPACITOR” or “RESERVOIR CAPACITOR”. Even after using this capacitor a small amount of ripple will remain.

We place the Filter Capacitor at the output of the rectifier the capacitor will charge to the peak voltage during each half cycle then will discharge its stored energy slowly through the load while the rectified voltage drops to zero, thus trying to keep the voltage as constant as possible.

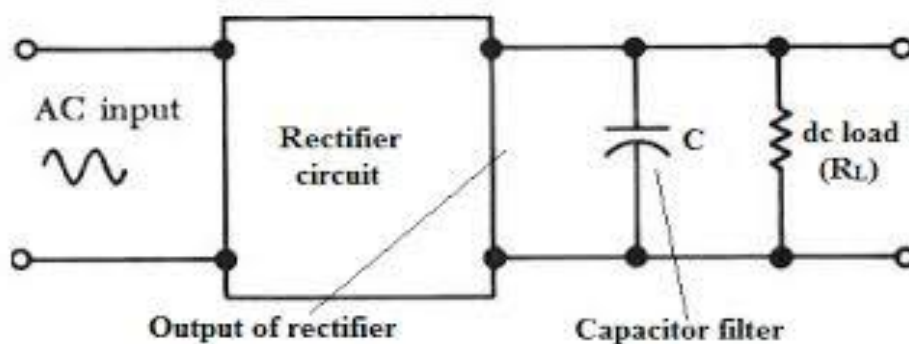


Figure 4.5: Filter Capacitor Circuit

If we go on increasing the value of the filter capacitor then the Ripple will decrease. But then the costing will increase. The value of the Filter capacitor depends on the current Consumed by the circuit, the frequency of the waveform & the accepted ripple.

$$C = \frac{V_r F}{I}$$

Where,

V_r = accepted ripple voltage.(should not be more than 10% of the voltage)

I = current consumed by the circuit in Amperes.

F = frequency of the waveform. A half wave rectifier has only one peak in one cycle so $F=25\text{hz}$

Whereas a full wave rectifier has Two peaks in one cycle so $F=100\text{hz}$.

Voltage Regulator

A Voltage regulator is a device which converts varying input voltage into a constant regulated output voltage. Voltage regulator can be of two types

1) Linear Voltage Regulator

Also called as Resistive Voltage regulator because they dissipate the excessive voltage resistively as heat.

2) Switching Regulators.

They regulate the output voltage by switching the Current ON/OFF very rapidly. Since their output is either ON or OFF it dissipates very low power thus achieving higher efficiency as compared to linear voltage regulators. But they are more complex & generate high noise due to their switching action. For low level of output power switching regulators tend to be costly but for higher output wattage they are much cheaper than linear regulators.

After filtering the rectifier output the signal is given to a voltage regulator. The maximum input voltage that can be applied at the input is 35V.

Circuit diagram

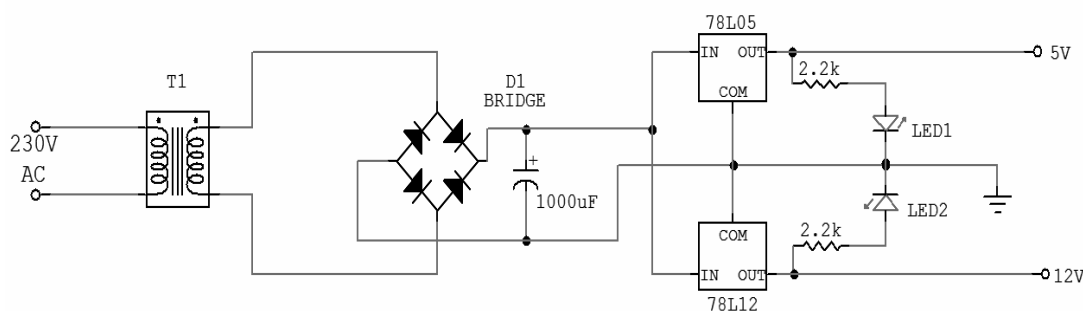


Figure 4.6: Voltage Regulator Circuit

LCD Display

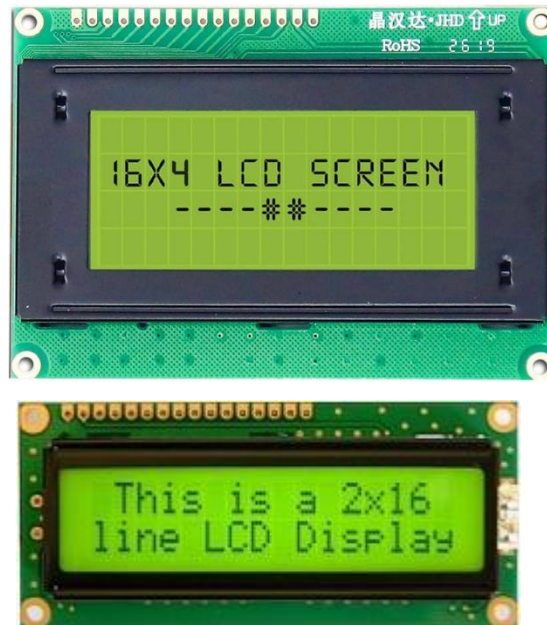


Figure 4.7: LCD Display Module

LCD Interfacing

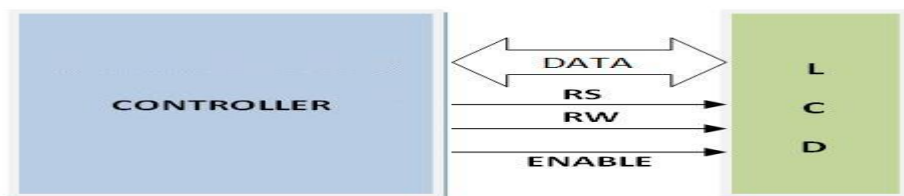


Figure 4.8: LCD Interfacing Diagram

SOIL SENSOR

This moisture sensor can read the amount of moisture present in the soil surrounding it. It's a low-tech sensor, but ideal for monitoring an urban garden, or your pet plant's water level. This is a must have tool for a connected garden!



Figure 4.9: Soil Moisture Sensor

DHT-11

This board is a breakout board for the DHT11 sensor and gives a digital output that is proportional to temperature and humidity measured by the sensor. Technology used to produce the DHT11 sensor grants high reliability, excellent long-term stability and very fast response time.

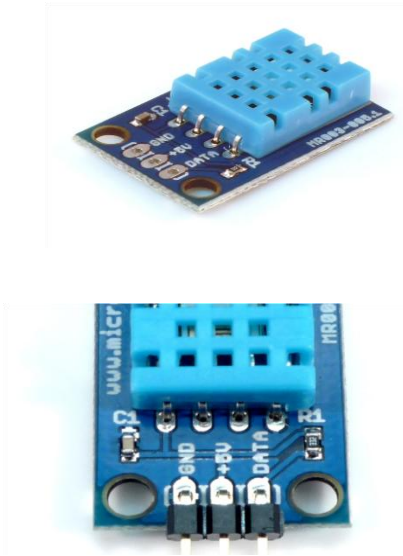


Figure 4.10: DHT11 Temperature and Humidity Sensor

RELAY(1-CHANNEL)



Figure 4.11: Relay Module Circuit

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The system architecture of the Smart Agriculture Monitoring and Controlling System is designed as a structured flow consisting of sensing, processing, and actuation layers. Sensors such as the DHT11 and soil moisture sensor form the input layer, continuously collecting real-time environmental data like temperature, humidity, and soil moisture. This data is sent to the Arduino Uno microcontroller, which acts as the processing unit, where it is analyzed based on predefined threshold values. The I2C LCD display provides real-time visual feedback to the user, while a relay module controls devices such as the water pump motor. If the soil moisture level falls below the set limit, the system automatically activates irrigation, and once the desired level is reached, it turns the pump off. Thus, the architecture follows a simple and efficient “sense–process–act” model, ensuring automated operation, reduced manual effort, and efficient resource utilization, with the flexibility to integrate wireless communication for remote monitoring in future enhancements.

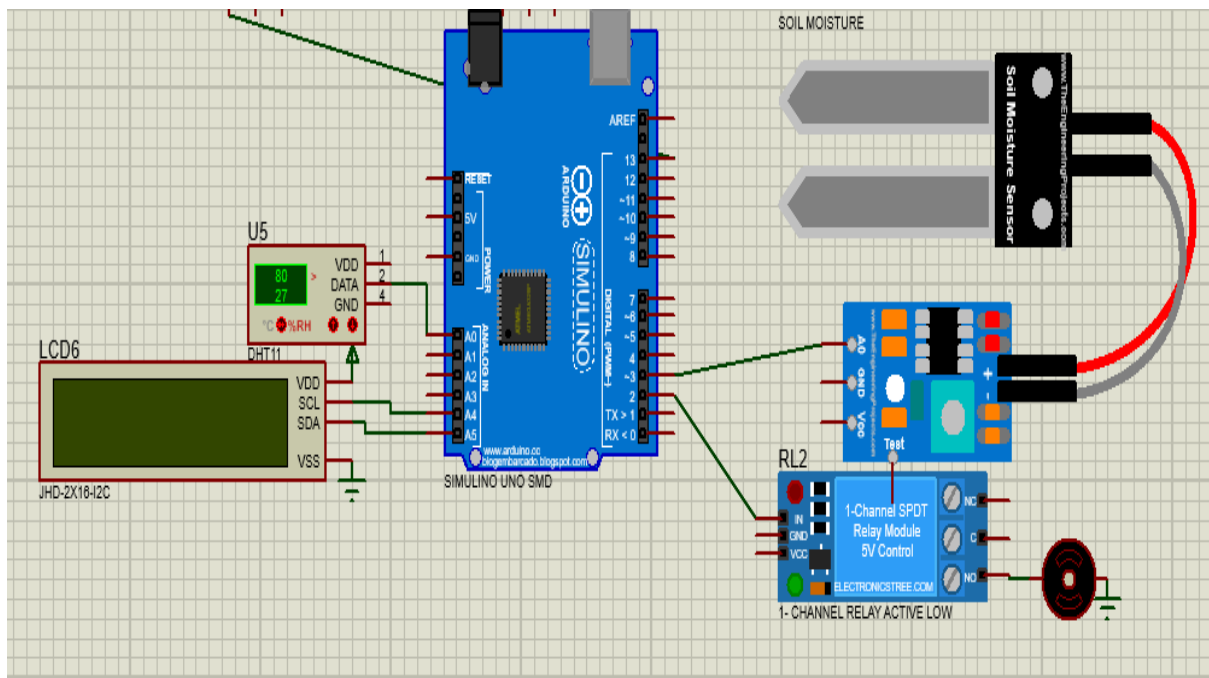


Figure 5.1: Smart Irrigation System Architecture

5.2 Source Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>
// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);

// DHT
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

#define WATER_PIN 3
#define MOTOR 5

void setup() {
  pinMode(WATER_PIN, INPUT);
  pinMode(MOTOR, OUTPUT);

  lcd.init();
  lcd.backlight();
  dht.begin();

  lcd.setCursor(0, 0);
  lcd.print("SMART IRRIGATION");
  lcd.setCursor(2, 1);
  lcd.print("INITIALIZING");
  delay(3000);
  lcd.clear();
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
```

```

int waterState = digitalRead(WATER_PIN);
String SoilStatus = (waterState == LOW) ? "WET" : "DRY";
bool waterDetected = (waterState == HIGH);

digitalWrite(MOTOR, waterDetected);

// LCD DISPLAY
lcd.setCursor(0, 1);
lcd.print("Temp:");
lcd.print((int)temp);
lcd.write(0xDF);
lcd.print("C ");

lcd.setCursor(0, 0);
lcd.print("SOIL:");
lcd.print(SoilStatus);

lcd.print(" Hum:");
lcd.print((int)hum);
lcd.print("%");

delay(1000);
}

```

CHAPTER 6

SOFTWARE IMPLEMENTATION

6.1 Software Development

These are the steps you need to follow in order to be up and running:

- Get an ARDUINO UNO board
- Download the ARDUINO UNO environment
- Install the USB drivers
- Connect the board
- Upload a program

Get an ARDUINO UNO board

The ARDUINO UNO i/o board is a simple circuit featuring the ATmega8 processor from Atmel. The board is composed of a printed circuit board (PCB) and electronic parts.

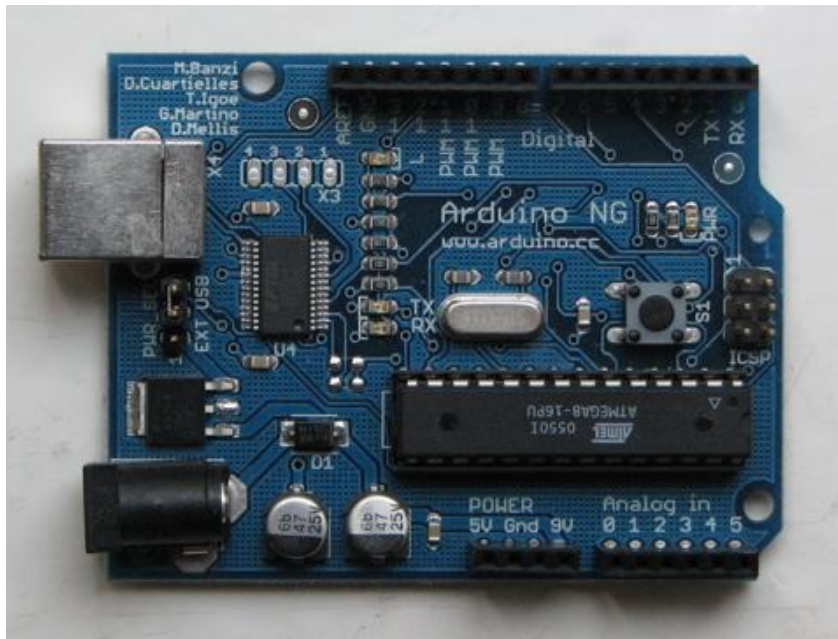


Figure 6.1: Arduino IDE Program Output

There are a few ways to get an ARDUINO UNO board:

buy a ready-made board. See how to buy a board or just the PCB.

European distributor

US distributor

Build the own board. Also build the own PCB just by downloading the CAD files from the Hardware page. Extract the .brd file and send it to a PCB manufacturer. Be aware that manufacturing a single pcb will be very expensive. It's better to get together with other people and make 20 or 30 at a time. Since you get the full CAD files to make the own

customised version of ARDUINO UNO. if you make modifications or fix bugs please send us your changes!

Purchase parts.

Purchase the parts from any electronics store. The Serial version in particular has been designed to use the most basic parts that can be found anywhere in the world. The USB version on the other hand requires some advanced soldering skills because of the FTDI chip that is a smd part.

Assemble the board.

Put together a step by step guide on how to build an ARDUINO UNO board. Newbies: never soldered before? afraid of trashing thousands of boards before getting one properly soldered? fear not learn to master the art of soldering.

Program the boot loader.

In order for the development environment to be able to program the chip, this has to be programmed with a piece of code called bootloader. See the bootloader page on how to program it on your chip.

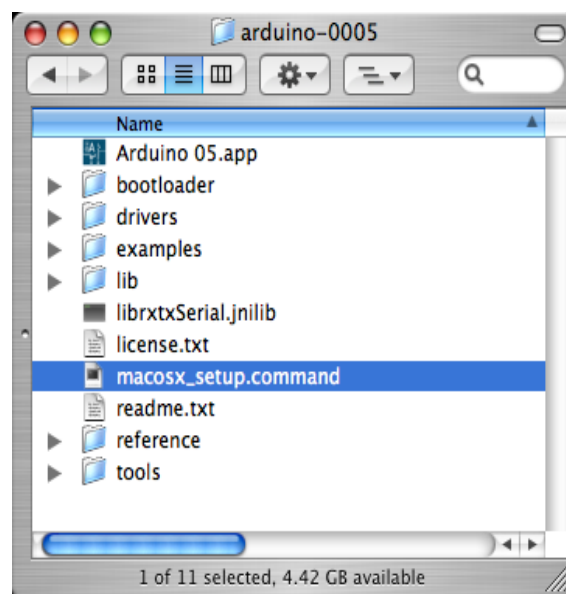
Download the ARDUINO UNO environment

To program the ARDUINO UNO board, you need the ARDUINO UNO environment.

Download ARDUINO UNO: From the software page.

Linux note: For help getting the ARDUINO UNO IDE running on Debian, please see the FAQ ("How can I run the ARDUINO UNO IDE under Linux?").

Mac OS X note: After downloading the IDE, run the macosx_setup.command. It corrects permission on a few files for use with the serial port and will prompt you for your password. You may need to reboot after running this script.



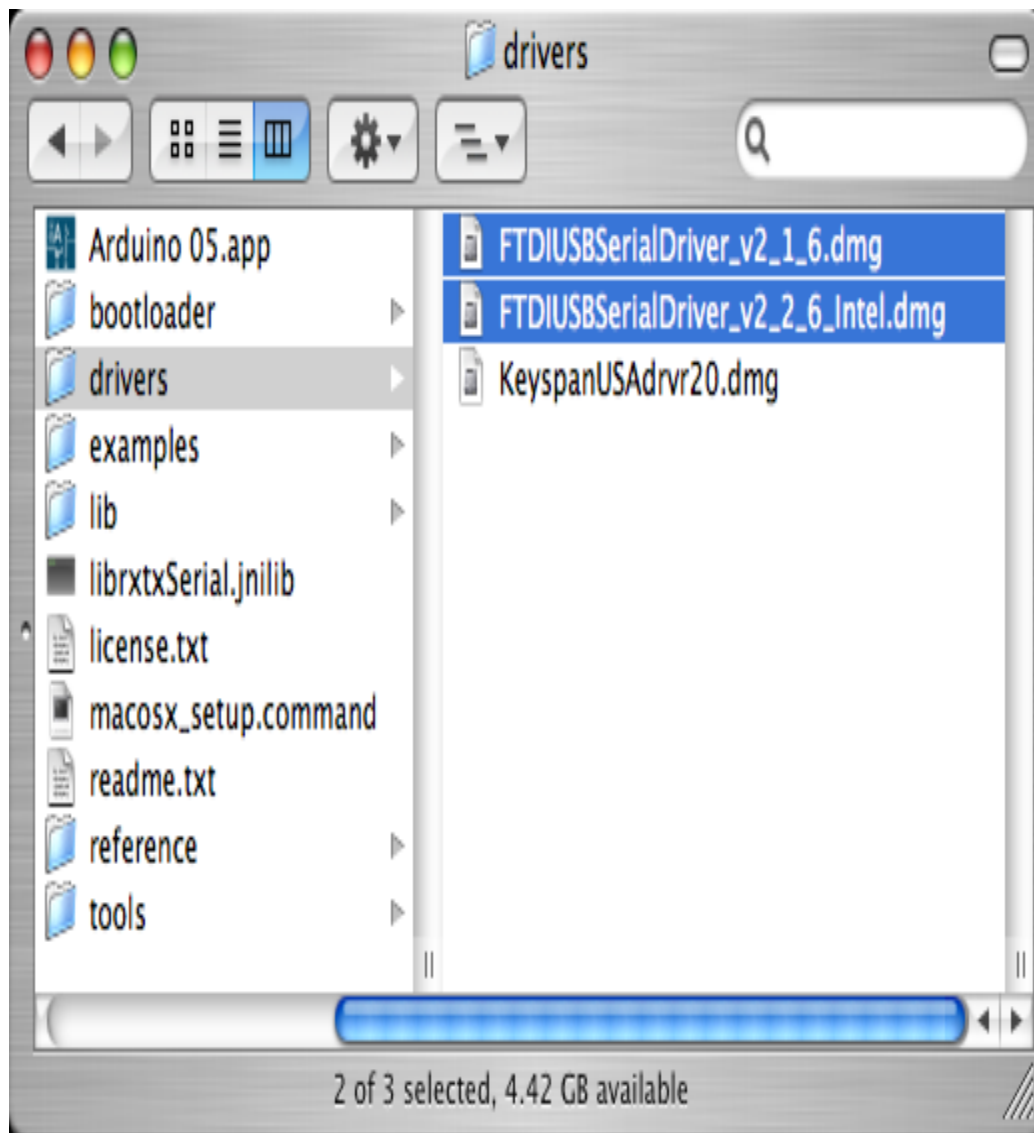
Install the USB drivers

If you are using a USB ARDUINO UNO, you will need to install the drivers for the FTDI chip on the board.

These can be found in the driver's directory of the ARDUINO UNO distribution.

On Windows, you will need to unzip FTDI USB Drivers.zip. Then, when you plug in the ARDUINO UNO board, point the Windows Add Hardware wizard to the FTDI USB Drivers directory.

On the Mac, mount the FTDIUSBSerialDriver_v2_1_6.dmg (on PPC machines) or the FTDIUSBSerialDriver_v2_2_6_Intel.dmg (on Intel machines) disk image and run the included FTDIUSBSerialDriver.pkg.



The latest version of the drivers can be found on the FTDI website.

Connect the board

If you're using a serial board, power the board with an external power supply (6 to 25 volts DC, with the core of the connector positive). Connect the board to a serial port on your computer.

On the USB boards, the power source is selected by the jumper between the USB and power plugs. To power the board from the USB port (good for controlling low power devices like LEDs), place the jumper on the two pins closest to the USB plug.

To power the board from an external power supply (needed for motors and other high current devices), place the jumper on the two pins closest to the power plug. Either way, connect the board to a USB port on your computer. On Windows, the Add New Hardware wizard will open; tell it you want to specify the location to search for drivers and point to the folder containing the USB drivers you unzipped in the previous step.

The power LED should go on.

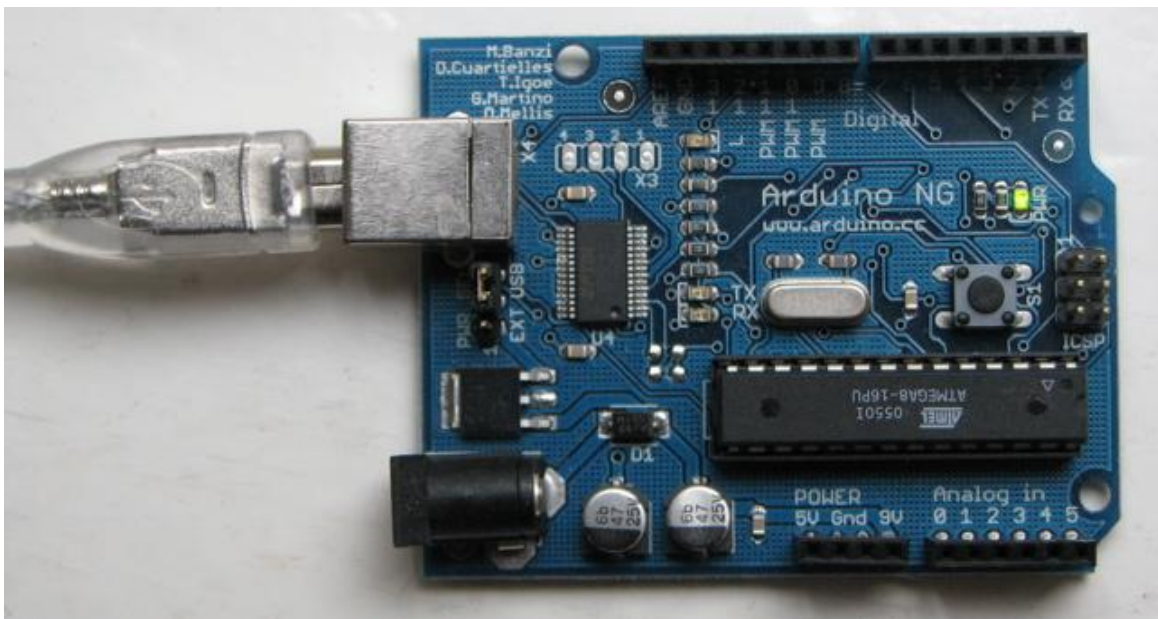
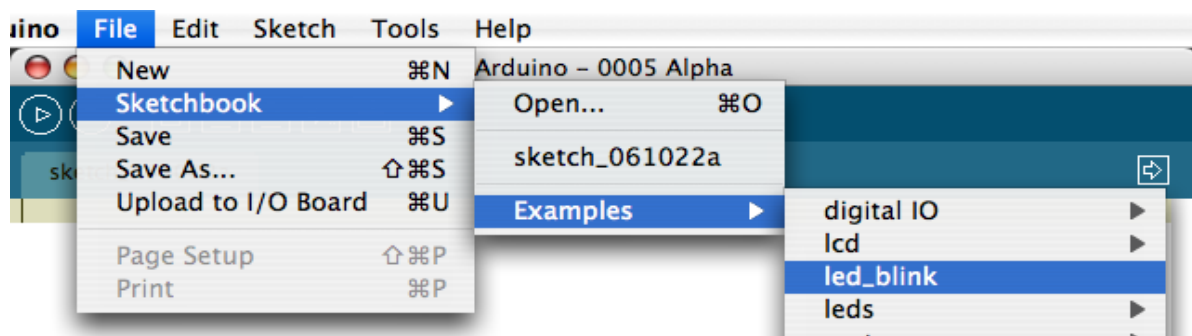


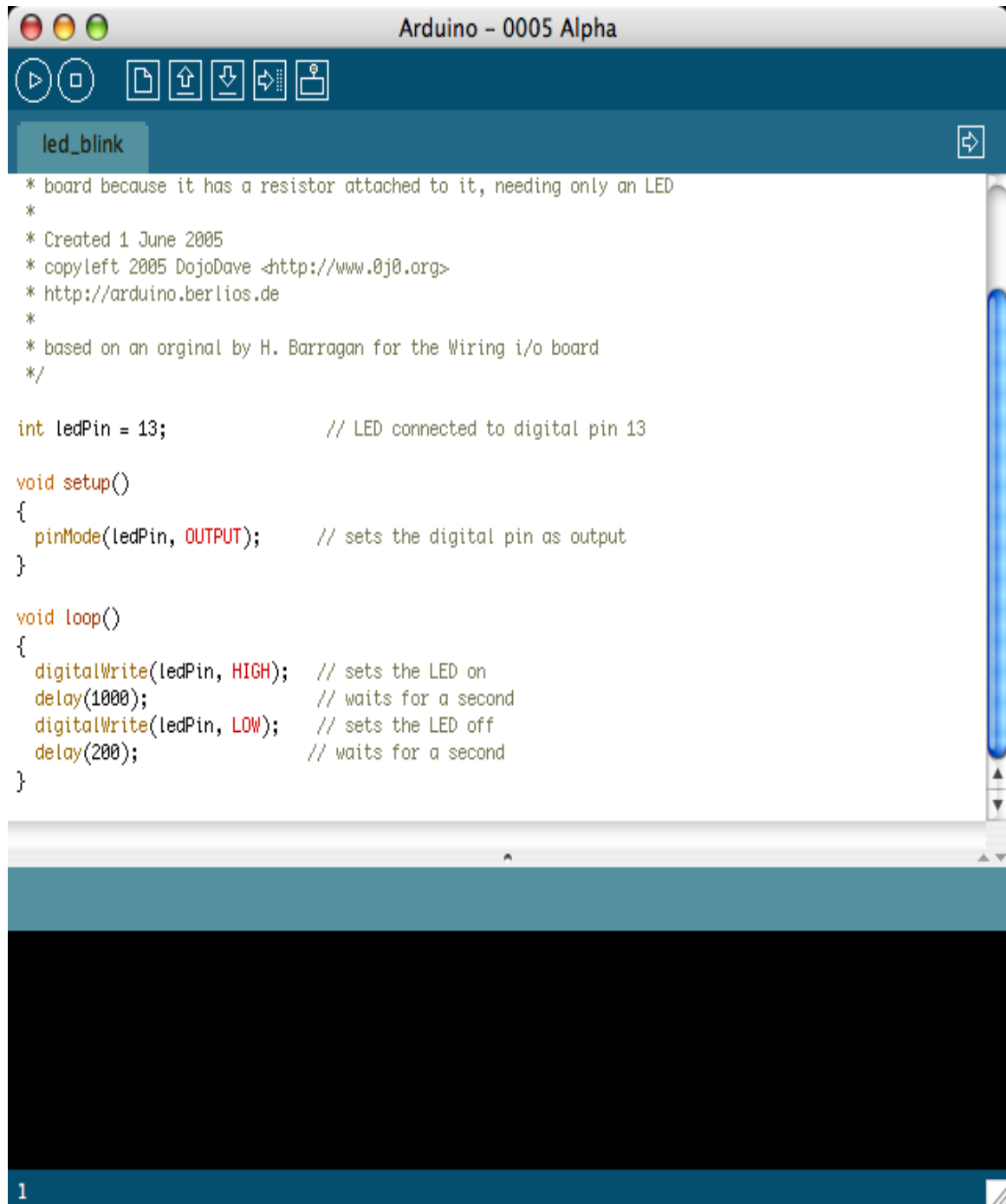
Figure 6.2: LED Blink Program Output

Upload a program

Open the LED blink example sketch: File > Sketchbook > Examples > led blink.



Here's what the code for the LED blink example looks like.

A screenshot of the Arduino IDE window titled "Arduino - 0005 Alpha". The window has a dark blue header bar with icons for running, stopping, saving, and other functions. Below the header, the file name "led_blink" is shown in a light blue tab. The main text area contains C++ code for an LED blink. The code includes comments about the board's resistor, creation date (1 June 2005), copyright (2005 DojoDave), and a link to the Arduino website. It defines a pin number (13) and contains two functions: setup() which configures the pin as an output, and loop() which turns the LED on and off with delays. The status bar at the bottom shows the line number "1".

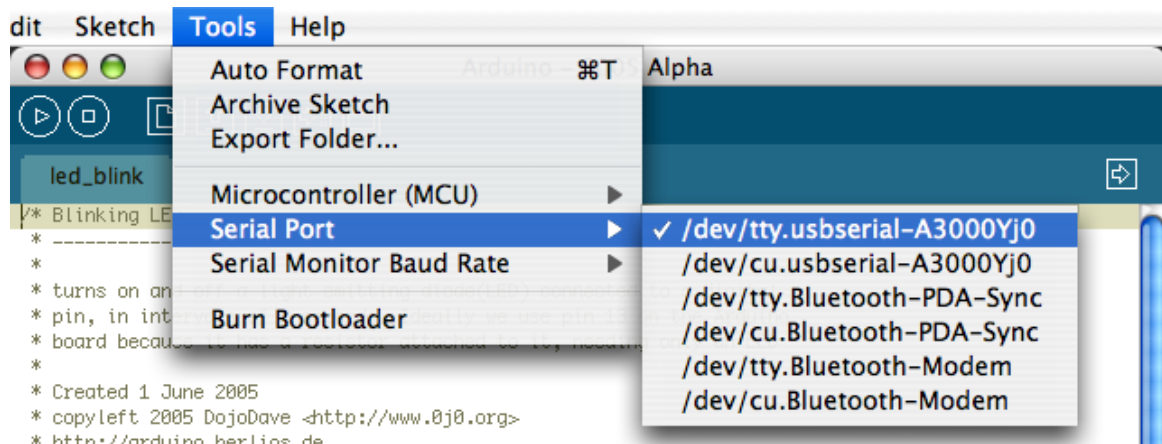
```
* board because it has a resistor attached to it, needing only an LED
*
* Created 1 June 2005
* copyleft 2005 DojoDave <http://www.0j0.org>
* http://arduino.berlios.de
*
* based on an original by H. Barragan for the Wiring i/o board
*/

int ledPin = 13;          // LED connected to digital pin 13

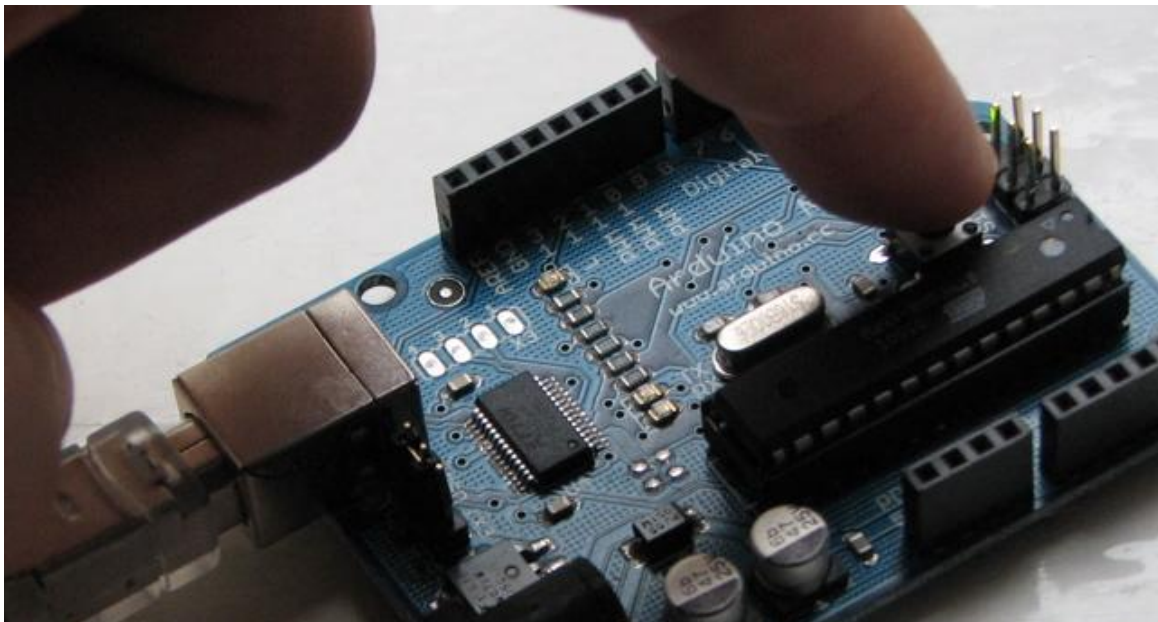
void setup()
{
  pinMode(ledPin, OUTPUT); // sets the digital pin as output
}

void loop()
{
  digitalWrite(ledPin, HIGH); // sets the LED on
  delay(1000);                // waits for a second
  digitalWrite(ledPin, LOW);  // sets the LED off
  delay(200);                 // waits for a second
}
```

Select the serial device of the ARDUINO UNO board from the Tools | Serial Port menu. On Windows, this should be COM1 or COM2 for a serial ARDUINO UNO board, or COM3, COM4, or COM5 for a USB board. On the Mac, this should be something like /dev/cu.usbserial-1B1 for a USB board, or something like /dev/cu. USA19QW1b1P1.1 if using a Keyspan adapter with a serial board (other USB-to-serial adapters use different names).



Push the reset button on the board then click the Upload button in the IDE. Wait a few seconds. If successful, the message "Done uploading." will appear in the status bar.



If the ARDUINO UNO board doesn't show up in the Tools | Serial Port menu, or you get an error while uploading, please see the FAQ for troubleshooting suggestions.

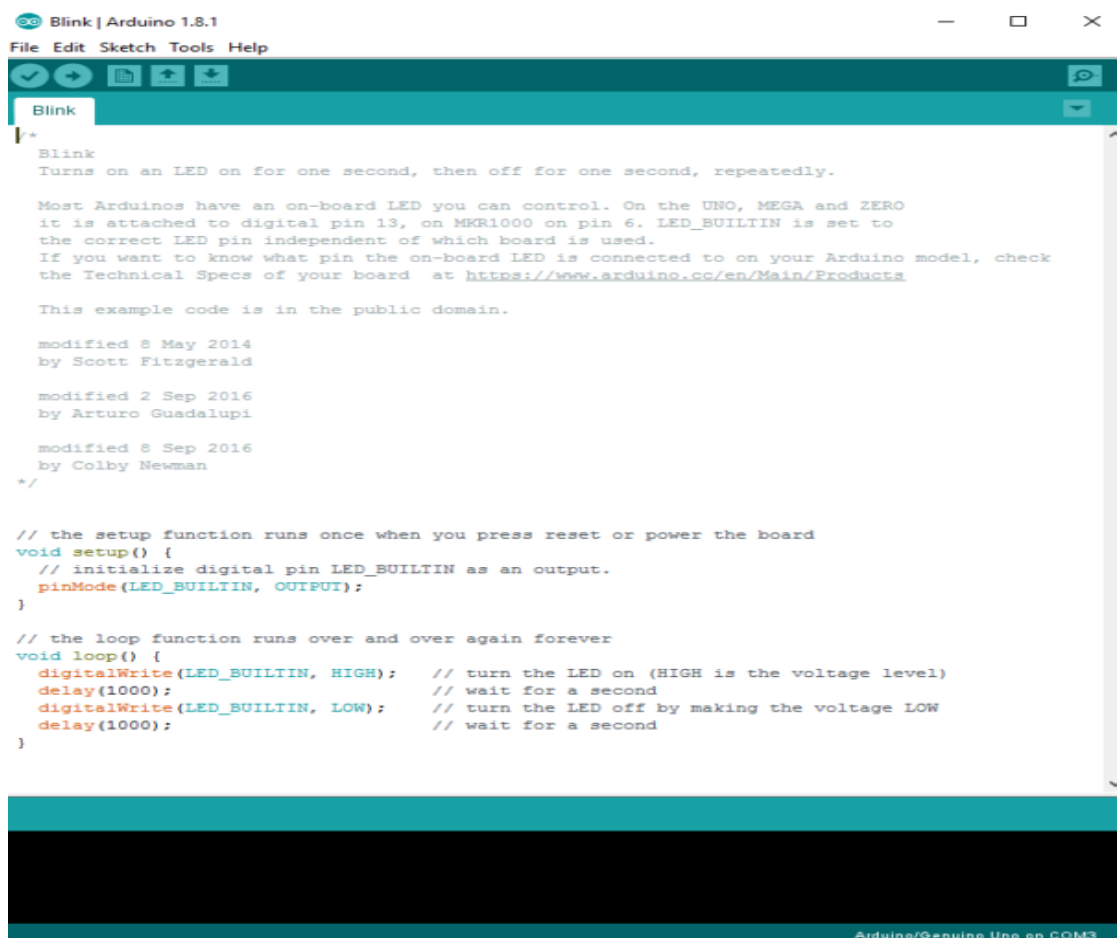
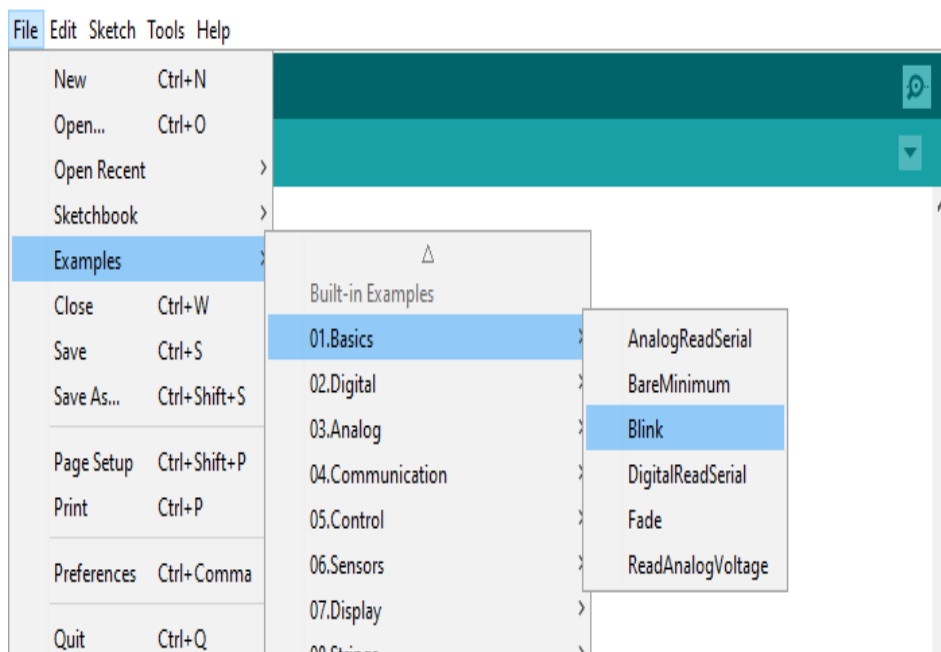
A few seconds after the upload finishes, you should see the amber (yellow) LED on the board start to blink.

Example Programs

To open the blink sketch, you will need to go to

File

- >Examples
- > Basics
- > Blink



Now you should have a fully coded blink sketch that looks like the image below.

Next, you need to click on the verify button (check mark) that's located in the top left of the IDE box. This will compile the sketch and look for errors. Once it says "Done Compiling" you are ready to upload it. Click the upload button (forward arrow) to send the program to the ARDUINO UNO board.



The built-in LEDs on the ARDUINO UNO board will flash rapidly for a few seconds and then the program will execute. If everything went correctly, the LED on the breadboard should turn on for a second and then off for a second and continue in a loop.

Change the Code

Before going to the next project, let's change some of the code in the "Blink" sketch to make it do something different. Playing around with the sketch will help you start to learn how the code controls the board.

Keep the ARDUINO UNO board connected and change the delay portion of the code from (1000) to (200). Click the verify button on the top left of the IDE and then click upload. This should make the LED on the breadboard blink faster.

NOTE – ARDUINO UNO measures time in milliseconds and 1000 milliseconds = 1 second. The original code (1000) turns on the LED for 1 second and then off for 1 second. By adjusting the code from (1000) to (200) it shortens the time between on and off which makes it blink faster.

```
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);
  delay(200);
  digitalWrite(LED_BUILTIN, LOW);
  delay(200);
}
```

Changed to 200

Changed to 200

ARDUINO UNO Project 2: LED w/ Switch

Now it's time to talk switches and how they can be incorporated into ARDUINO UNO projects. A switch is a electrical component that completes a circuit when pushed and breaks the circuit when released. In this project, we will be using a small pushbutton switch to control an LED.

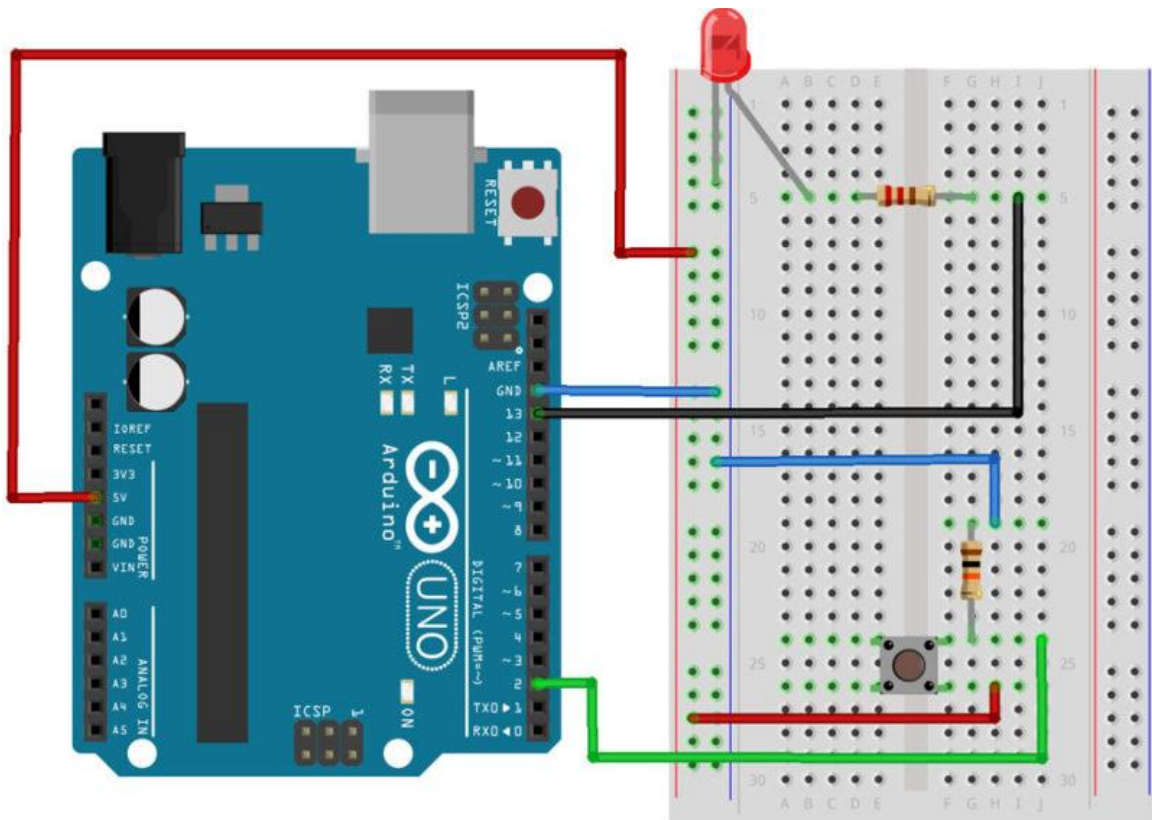


Figure 6.3: Switch Circuit Using Arduino

Required Parts

ARDUINO UNO Uno Board

Breadboard – half size

Jumper Wires

USB Cable

LED (5mm)

Push button switch

10k Ohm Resistor

220 Ohm Resistor

Connect the Parts

To build an ARDUINO UNO circuit by looking at the breadboard image above or by using the written description below. In the written description, Use a letter/number combo that refers to the location of the component. If H19 is mentioned for example, that refers to column H, row 19 on the breadboard.

Step 1 – Connect the blue jumper wire from the GND on the ARDUINO UNO to the GND rail (blue line) on the breadboard near A13

Step 2 – Connect the blue jumper wire from the GND rail on the breadboard near A17 to H19

Step 3 – Connect the red jumper wire from the power rail on the breadboard around row A27 to H26

Step 4 – Connect the green jumper wire from pin 2 on ARDUINO UNO to J24 on the breadboard

Step 5 – Place one leg of a 10k Ohm resistor in G19 and the other leg in G24

Step 6 – Place the pushbutton switch into F24, F26, E24 and E26

Step 7 – Place one leg of a 220 Ohm resistor in D5 and the other leg in G5

Step 8 – Insert the short leg of the LED in the GND rail around A5 and the long leg in B5

Step 9 – Connect the black jumper wire from pin 13 on the ARDUINO UNO to I5 on the breadboard

Step 10 – Connect the red jumper wire from 5V on the ARDUINO UNO to power rail (+) near A8

Step 11 – Connect the ARDUINO UNO Uno to your computer via USB cable

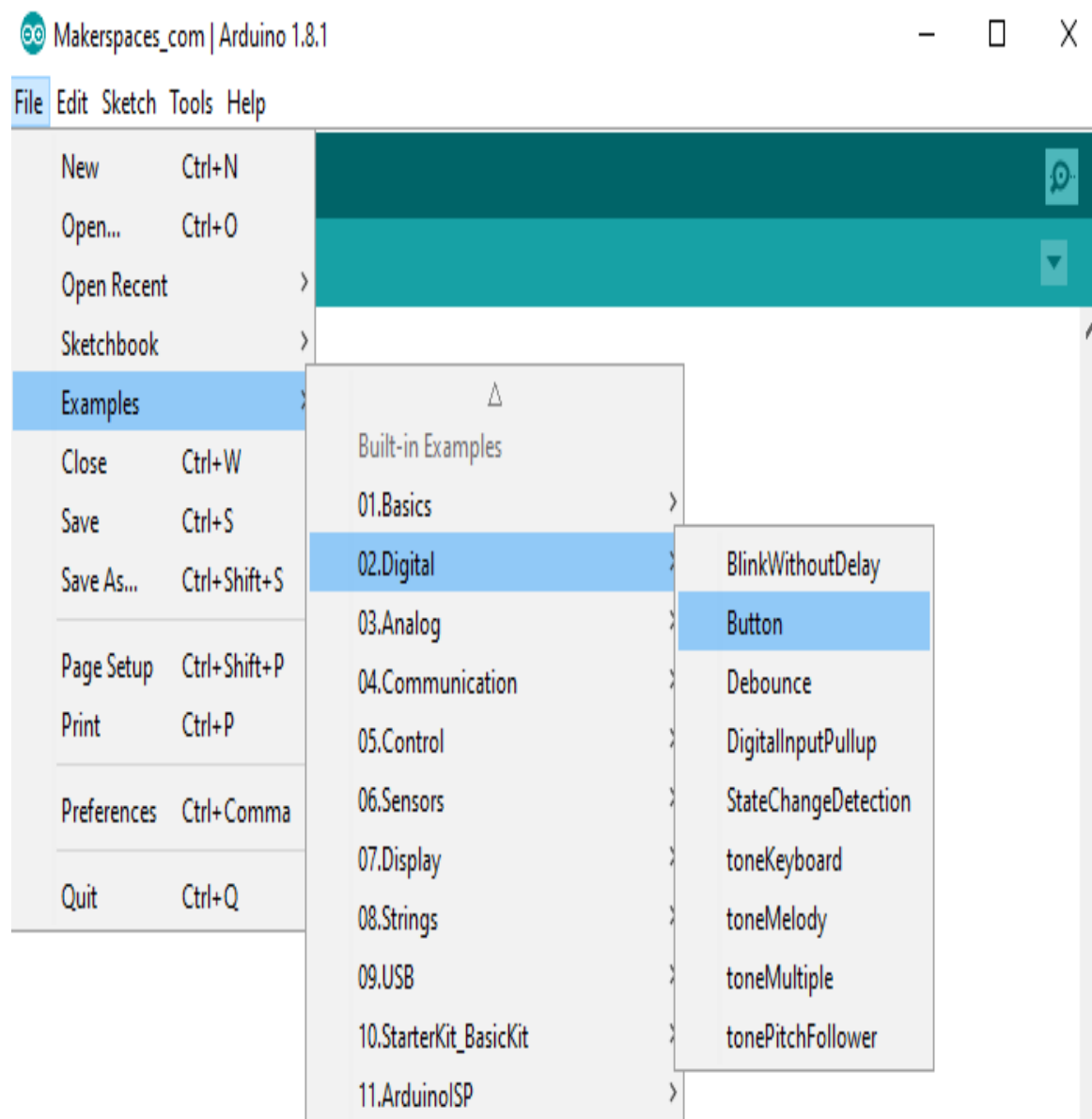
Upload the Switch Sketch

Now it's time to upload the sketch to the ARDUINO UNO that will allow to use a switch. As with the blink sketch, there are example programs already loaded in the ARDUINO UNO IDE that we will be using.

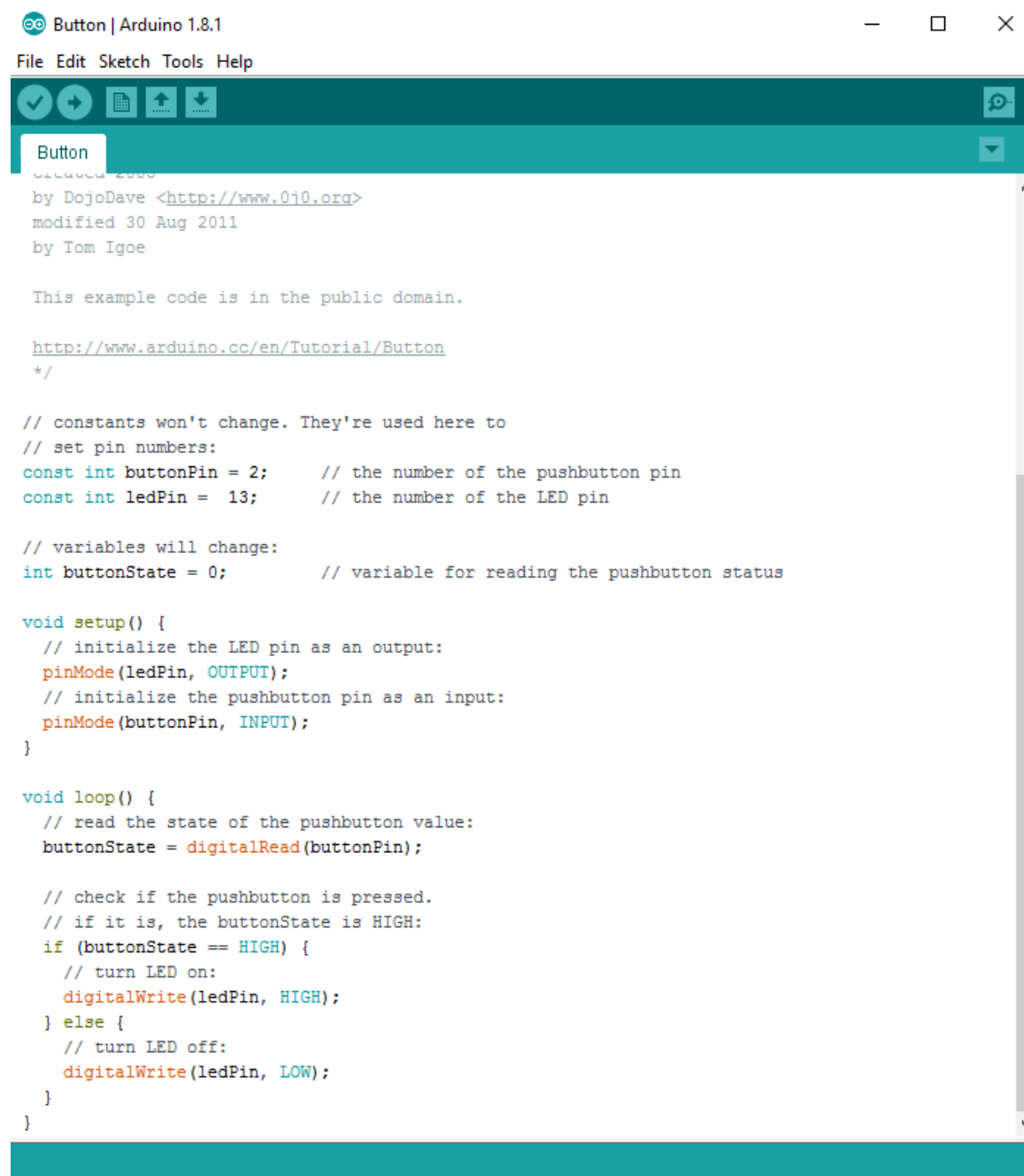
In order to use a switch, Load the file called "Button" which can be found here:

File

- > Examples
- > Digital
- > Button



Now Check for a fully coded button sketch that looks like the image below.

A screenshot of the Arduino IDE interface. The title bar reads "Button | Arduino 1.8.1". The menu bar includes "File", "Edit", "Sketch", "Tools", and "Help". Below the menu bar is a toolbar with icons for checking, uploading, and other functions. The main text area contains the following code:

```
Button
Created: 2009
by DojoDave <http://www.0j0.org>
modified 30 Aug 2011
by Tom Igoe

This example code is in the public domain.

http://www.arduino.cc/en/Tutorial/Button
*/

// constants won't change. They're used here to
// set pin numbers:
const int buttonPin = 2;    // the number of the pushbutton pin
const int ledPin = 13;      // the number of the LED pin

// variables will change:
int buttonState = 0;        // variable for reading the pushbutton status

void setup() {
  // initialize the LED pin as an output:
  pinMode(ledPin, OUTPUT);
  // initialize the pushbutton pin as an input:
  pinMode(buttonPin, INPUT);
}

void loop() {
  // read the state of the pushbutton value:
  buttonState = digitalRead(buttonPin);

  // check if the pushbutton is pressed.
  // if it is, the buttonState is HIGH:
  if (buttonState == HIGH) {
    // turn LED on:
    digitalWrite(ledPin, HIGH);
  } else {
    // turn LED off:
    digitalWrite(ledPin, LOW);
  }
}
```

Next, Click on the verify button (check mark) that's located in the top left of the IDE box. Once it says "Done Compiling" get ready to upload it. Click the upload button (forward arrow) to send the program to the ARDUINO UNO board.

CHAPTER 7

SYSTEM TESTING

Testing is an important phase in system development to ensure that the Smart Irrigation System works correctly, efficiently, and without errors. Different levels of testing are performed to verify both hardware and software components.

7.1 Unit Testing

Unit testing is an essential step in the development of the Smart Agriculture Monitoring and Controlling System, where each individual component is tested separately to ensure proper functionality before integrating the entire system. In this process, components such as the soil moisture sensor, DHT-11 sensor, relay module, LCD display, and Arduino code are verified independently. For instance, the soil moisture sensor is tested to check whether it provides accurate readings based on different soil conditions, while the DHT-11 sensor is evaluated to ensure it correctly measures and displays temperature and humidity values. Similarly, the relay module is tested to confirm that it switches ON and OFF as expected when triggered by the Arduino. The LCD display is also checked to verify that it shows the correct data clearly. Additionally, the Arduino code is debugged and validated to ensure proper logic execution and communication between all components.

7.2 Integration Testing

Integration testing is performed to ensure that all components of the Smart Agriculture Monitoring and Controlling System work together seamlessly as a complete unit. In this phase, individual modules such as sensors, Arduino, relay, and LCD display are interconnected and tested collectively to verify proper communication and data flow. The soil moisture sensor and DHT-11 sensor are connected to the Arduino, which processes the incoming data and sends appropriate signals to the relay module and LCD display. The LCD continuously shows updated readings, confirming that data is being correctly transmitted and processed. For example, when the soil moisture level drops below the set threshold, the sensor sends this data to the Arduino, which then processes it and activates the relay module, resulting in the water pump turning ON automatically. Similarly, when the desired moisture level is reached, the system turns the pump OFF.

7.3 System Testing

System testing is carried out to evaluate the performance of the entire Smart Agriculture Monitoring and Controlling System as a complete and integrated unit under real-time conditions. In this phase, all components—including sensors, Arduino, relay module, water pump, and LCD display—are tested together to ensure the system functions according to the project requirements. The testing focuses on key aspects such as the accuracy of the displayed sensor data, the effectiveness of automatic irrigation control, and the response time of the system when environmental conditions change. For example, when the soil becomes dry, the system should correctly detect the low moisture level, process the data through the Arduino, and automatically activate the relay to turn the water pump ON. Similarly, when the soil reaches an adequate moisture level, the system should promptly turn the pump OFF. The LCD display is also monitored to ensure it shows accurate and updated readings.

7.4 Acceptance Testing

Acceptance testing is conducted to ensure that the Smart Agriculture Monitoring and Controlling System meets the expectations and requirements of end users, such as farmers or project evaluators. In this phase, the system is tested in a real-world environment by users to evaluate its usability, accuracy, and reliability. The focus is on how easily the system can be operated, whether the sensor readings and automation are accurate, and how consistently the system performs over time. For example, a farmer may observe whether the system effectively reduces manual effort by automatically controlling irrigation based on soil moisture levels, without requiring constant monitoring. The user also checks if the LCD display provides clear and understandable information and whether the system responds correctly to changing environmental conditions.

7.5 Test Cases for Online System

Test Case ID	Test Description	Input Condition	Expected Output	Actual Output	Result
TC1	Soil moisture detection	Dry soil	Pump should turn ON	Pump turned ON successfully	Pass

TC2	Soil moisture detection	Wet soil	Pump should turn OFF	Pump turned OFF successfully	Pass
TC3	Temperature sensing	High temperature	Display shows correct value	Temperature displayed correctly on LCD	Pass
TC4	Humidity sensing	Normal humidity	LCD displays humidity	Humidity value displayed correctly	Pass
TC5	Relay operation	Signal from Arduino	Relay switches correctly	Relay activated/deactivated as expected	Pass
TC6	LCD display	Sensor data input	Data displayed properly	All sensor values shown correctly on LCD	Pass
TC7	Full system test	All components connected	System works automatically	System operated automatically without errors	Pass

CHAPTER 8

RESULTS AND OUTPUT SCREENS

8.1 Initial User Interface

When the system is powered ON, the LCD screen displays an initialization message, indicating the successful startup of the Smart Irrigation System. This stage is crucial as it confirms that all hardware components are functioning properly and the system is preparing for operation. The appearance of the message “SMART IRRIGATION INITIALIZING” on the I2C LCD ensures that the Arduino Uno microcontroller has been powered successfully and has begun executing the programmed instructions. Simultaneously, all connected sensors, including the DHT11 sensor for temperature and humidity measurement and the soil moisture sensor, start initializing and stabilizing to provide accurate readings. During this brief period, the system performs internal checks and establishes communication between the microcontroller, sensors, relay module, and display unit. This initialization phase assures the user that the system is active, reliable, and ready to monitor environmental conditions in real time. It also helps in identifying any startup issues, as the absence of this message may indicate a problem with power supply or component connections. Overall, the initialization display acts as a confirmation step, ensuring that the smart irrigation system is fully prepared to carry out automated monitoring and control operations efficiently.



Figure 8.1: System Initialization Display

8.2 System Setup

The complete hardware setup of the Smart Agriculture Monitoring and Controlling System consists of all essential components integrated and connected to ensure smooth and efficient operation. At the core of the setup is the Arduino Uno microcontroller, which acts as the central unit responsible for processing data and controlling the entire system. The soil moisture sensor is inserted into the soil to continuously measure the moisture level, while the DHT-11 sensor is positioned to monitor environmental parameters such as temperature and humidity. These sensors are connected to the Arduino, which collects and analyzes the data in real time. An I2C LCD display is also interfaced with the Arduino to provide a clear and user-friendly interface, displaying current readings and system status. The relay module is used as a switching device to control the water pump, which is responsible for irrigation. When the soil moisture level falls below the predefined threshold, the Arduino sends a signal to the relay to activate the water pump, ensuring timely watering of crops. All components are properly wired using appropriate connections, ensuring stable communication and power distribution throughout the system. The entire setup is powered efficiently, and care is taken to maintain proper voltage levels to avoid any damage to components. This well-organized hardware configuration demonstrates the practical implementation of an automated irrigation system, highlighting its reliability, effectiveness, and suitability for real-world agricultural applications.

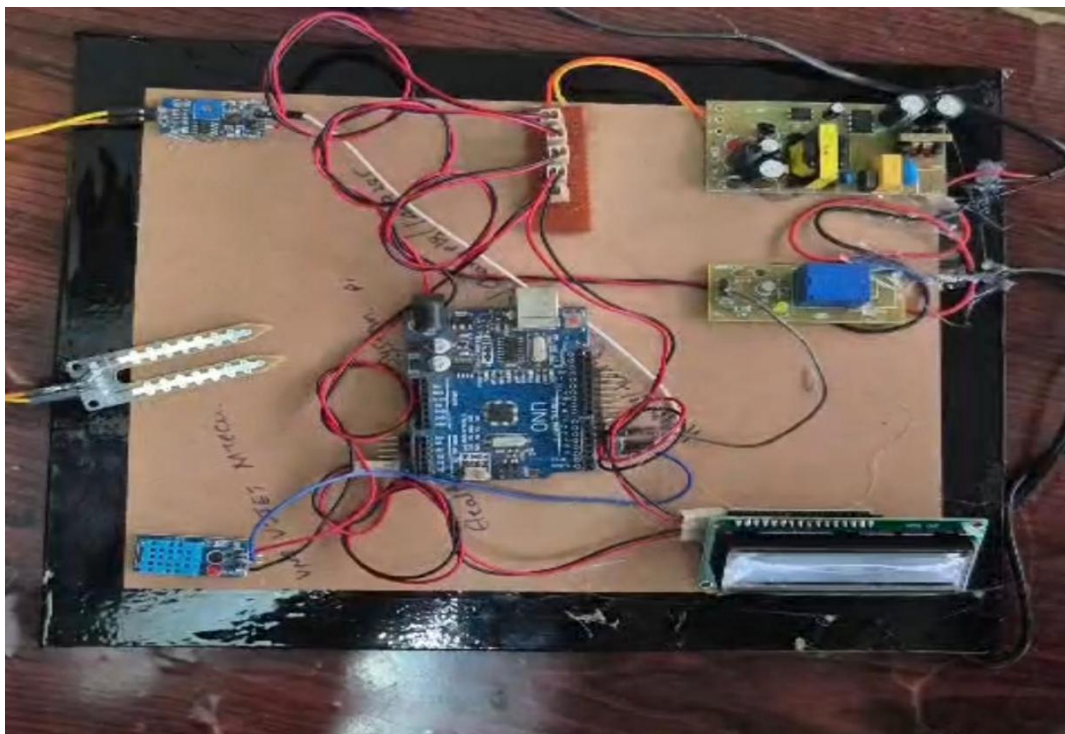


Figure 8.2: Complete Hardware Setup

8.3 Sensor Data Display

The Real-Time Sensor Data Display serves as the primary interface between the user and the Smart Agriculture Monitoring System, providing continuous updates about environmental conditions in the field. The I2C LCD screen dynamically shows important parameters such as soil condition, humidity, and temperature, allowing users to easily monitor the status of their crops. The soil moisture level is interpreted and displayed in a simplified form as “Wet” or “Dry,” enabling quick understanding without needing technical knowledge. Additionally, the humidity value (for example, 60%) and temperature reading (such as 29°C) are presented clearly, giving a complete overview of the surrounding environment. This real-time display ensures that farmers or users can make immediate decisions if required, although the system itself is automated. It eliminates the need for manual inspection of fields, saving time and effort while improving efficiency. The continuous monitoring also helps in detecting sudden environmental changes that may affect crop health. Overall, this display acts as a user-friendly dashboard, enhancing transparency, reliability, and ease of operation in the smart irrigation system, while supporting better crop management and precision agriculture practices.

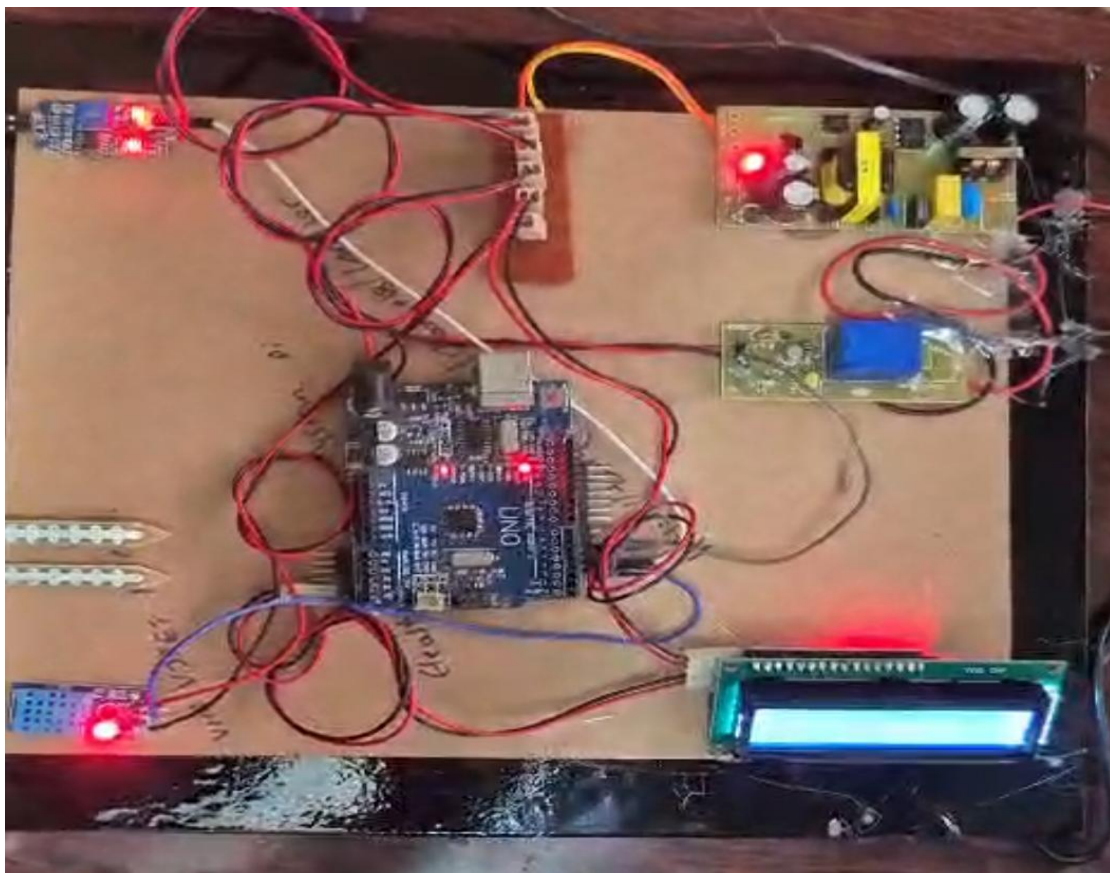


Figure 8.3: Real-Time Sensor Data Display

8.4 Final Output Display

The system dynamically displays updated environmental conditions on the LCD screen, providing continuous real-time feedback about the agricultural field. As sensor readings change, the display is automatically refreshed to show the latest values of temperature, humidity, and soil moisture, clearly indicating the current state of the environment. This real-time updating feature demonstrates that the system is actively monitoring conditions without any delay, ensuring accurate and timely information for the user. For instance, if there is a variation in temperature or a drop in soil moisture, the changes are instantly reflected on the display, confirming that the sensors and microcontroller are functioning correctly. This dynamic behavior not only enhances the reliability of the system but also builds user confidence in its performance. It eliminates the need for manual checking and allows users to observe how the system responds to environmental changes in real time. Overall, this feature highlights the efficiency and effectiveness of the smart agriculture system in providing continuous monitoring and supporting automated decision-making for better crop management.

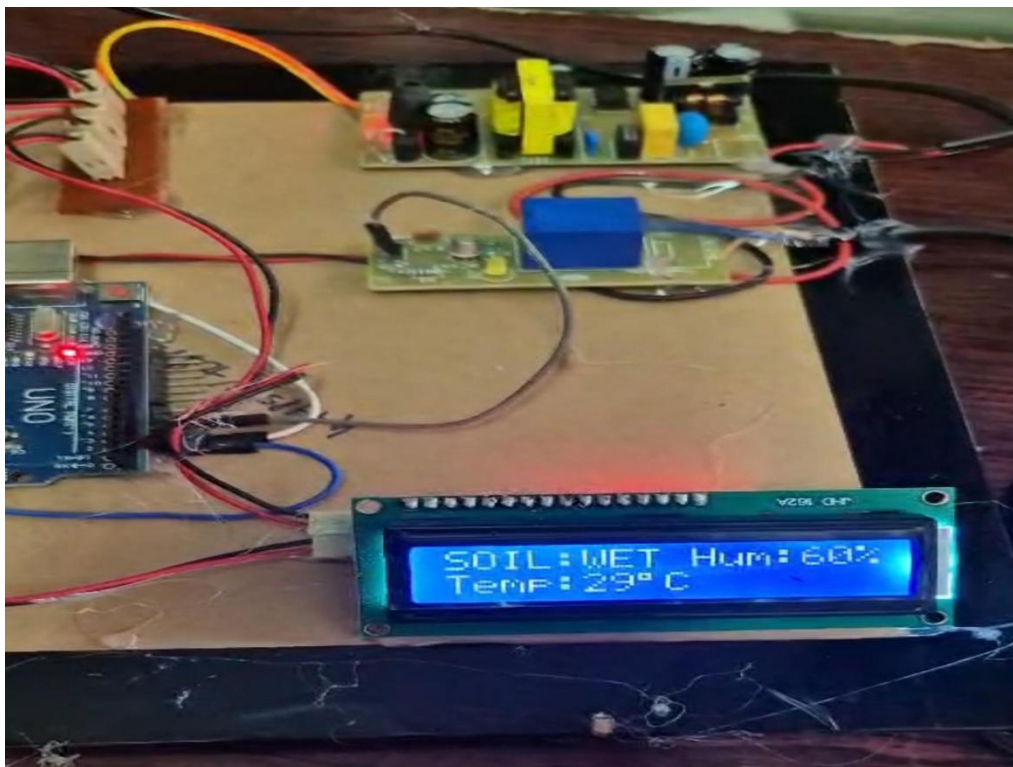


Figure 8.4: Final System Output Display

CHAPTER 9

CONCLUSION

In this project, we successfully developed and demonstrated a WSN-based Smart Agriculture Monitoring and Controlling System that effectively utilizes ARDUINO UNO, DHT-11 sensor, soil moisture sensor, relays crop management and automation. The system continuously monitors key environmental parameters such as temperature, humidity, and soil moisture, and responds automatically by triggering appropriate actuators—like activating a relay motor for irrigation when soil is dry and a relay-controlled system when the temperature exceeds the set threshold. The implementation of this system showcased the practicality and efficiency of using Wireless Sensor Networks to support sustainable and resource-efficient farming. By automating critical agricultural tasks, the system reduces the dependency on manual labour, conserves water, and ensures that crops receive the necessary environmental conditions for optimal growth. Moreover, the integration of a user-friendly LCD display provides farmers with real-time feedback, making the system scalable and adaptable for modern precision agriculture needs.

In addition to its current capabilities, the proposed system offers significant potential for future enhancements and real-world deployment. By integrating advanced technologies such as IoT connectivity, cloud storage, and mobile applications, farmers can remotely monitor and control agricultural operations from anywhere, further improving convenience and efficiency. The system can also be expanded by incorporating additional sensors for parameters like soil pH, light intensity, and nutrient levels, enabling more precise and data-driven farming decisions. Furthermore, the use of renewable energy sources such as solar power can make the system more sustainable and suitable for rural and remote areas with limited electricity access. Overall, this project lays a strong foundation for developing intelligent farming solutions that can adapt to evolving agricultural challenges and contribute to increased productivity, sustainability, and smart resource management.

CHAPTER 10

FUTURE ENHANCEMENTS

The Smart Irrigation System developed in this project provides an efficient and automated solution for irrigation using basic sensors and a microcontroller. However, there is significant scope for further improvement and expansion by integrating advanced technologies. The following enhancements can make the system more intelligent, user-friendly, and suitable for large-scale agricultural applications.

One of the major improvements is the integration of Internet of Things (IOT) technology. By connecting the system to the internet using modules like Wi-Fi (NodeMCU or ESP8266), farmers can monitor and control irrigation remotely through a mobile application or web interface. This will allow real-time tracking of soil moisture, temperature, and humidity from anywhere, making the system more convenient and efficient.

Another important enhancement is the implementation of a cloud-based database. Currently, the system does not store data, but with cloud integration, historical data can be recorded and analyzed. This helps farmers understand patterns in soil conditions and weather, enabling better decision-making for irrigation and crop management.

The system can also be improved by incorporating machine learning algorithms. These algorithms can analyze past data and predict optimal irrigation schedules based on weather forecasts, soil type, and crop requirements. This will make the system smarter and more adaptive, reducing water wastage even further.

Adding a mobile application is another useful enhancement. A simple Android app can provide notifications, alerts, and control options for the irrigation system. For example, the user can receive alerts when soil moisture is low or when the water pump is activated.

The use of solar power can make the system more sustainable and suitable for rural areas with limited electricity supply. Solar panels can be used to power the Arduino and other components, reducing dependency on external power sources.

Finally, the system can be scaled for large agricultural fields by using wireless sensor networks (WSN), allowing multiple sensor nodes to communicate with a central controller.

CHAPTER 11

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